One Sentence at a Time: A Quantitative History of Rationality in Economic Thought

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Abstract
This article demonstrates how unsupervised quantitative methods can enrich the history of economic thought. Using the largest English-language corpus ever assembled for the field—nearly 290,000 economics journal articles from 1900 to 2009 with citation data—we analyze the evolution of the concept of rationality. Combining large language model–based semantic analysis with bibliometric and network methods, we identify and cluster discussions of rationality across time and scales, such as the circulation of bounded rationality and the emergence of behavioral economics. We provide an open-source interactive tool to support transparency and reuse.

KEYWORDS
rationality; economics history; bibliometrics; behavioral economics; natural language processing; large language models

JEL CLASSIFICATION
B00; B20

Introduction

In the last decade, the number of history of economics papers employing quantitative methods has increased (see e.g., Goutsmedt, Claveau, and Herfeld 2023 special issue). Several essays have adopted a reflexive stance, examining the implications of using quantitative methods in the history of economics (Cherrier and Svorenčík 2018; Edwards, Giraud, and Schinckus 2018). These contributions provided broad discussions on the use of such methods. However, given the virtually unlimited variety of quantitative approaches potentially relevant to historians of economics—and the wide array of research questions they can address—these reflections often remain at an abstract level and offer little practical guidance. As a result, they tend to provide useful broad overviews, but without clearly articulating what quantification entails in practice, and how quantitative methods can be both meaningful and challenging to integrate into research specific to the history and philosophy of economics.

Our contribution sits between a historical study that uses quantitative methods to answer a specific question and a general methodological discussion of those methods. From the collection of data to the interpretation of results, we illustrate concretely how these methods are useful and examine, in practice, the methodological choices they entail. We present a step-by-step application of specific quantitative methods to a broad topic: the history of rationality in the twentieth century—using this case to illuminate, in practice, what choices quantitative analysis involves.

We focus our discussion on unsupervised methods. Unsupervised methods are machine-learning techniques in which algorithms identify patterns from unlabelled data, as opposed to supervised learning methods that learn patterns from labelled data to make inferences. The goal of unsupervised methods is not necessarily to provide a "measure" (Grimmer, Roberts, and Stewart 2022)—though they can be adapted to do so—but to organise in categories large corpora and enable accelerated and "distant reading" (Moretti 2013; Guldi 2023). They are well suited to historical inquiry because their unsupervised nature reduces the risk of presentism: rather than imposing present-day categories on the past, unsupervised algorithms uncover patterns directly from the data. When time is explicitly incorporated, researchers can map the discipline at different points of time, to identify the emergence and decline of subjects and concepts, and to assess the influence of specific economists or ideas.

Our article draws on two types of data, texts and citations. The analysis of textual corpora offers a direct window into the semantic content of debates, while citation data provides a means of tracing intellectual connections and channels of influence within the discipline. We show how a large corpus of documents can be classified based on the information contained in textual and citation data—what we call "*semantic clusters*" and "*bibliometric communities*."[1] Combining these two methods allows us both to illustrate concretely the use of two common approaches for the history of economics. This combination is also justified by what we regard as a crucial principle of quantitative analysis in history of economics in particular, and more broadly in the history of science: the triangulation of different sources of data and the validation of results through the comparison of findings obtained from distinct methods.

Above all, our discussion aims to illustrate a core principle for historical inquiry with such unsupervised quantitative methods. They require continuous back-and-forth between aggregate quantitative results, complementary indicators used for interpretation, and preliminary knowledge and close reading of primary and secondary sources. These methods function not only as a form of corroboration but also as "discovery methods" (Grimmer, Roberts, and Stewart 2022): they facilitate the exploration of large datasets to reveal historical patterns. They often confirm, and sometimes complement, established findings. They can also reveal pitfalls and blind spots in existing research.

The concept of "rationality" is an effective focus for this concrete demonstration. First, many historians of economics engage with it in one way or another, which makes the exercise relevant for a broad audience. Second, because the concept is broad and pervasive in economics, applying quantitative methods to a very large corpus is particularly informative.[2] We use a corpus of around 290 000 articles in economics between 1900 and 2009 to show how these methods can handle long time horizons. Third, the multiple meanings attached to "rationality" and its uses applied to various subjects show how textual methods, coupled with bibliometrics, can help capture this semantic plurality.

In what follows, we concentrate on what such a quantitative analysis requires in practice. We highlight what kinds of questions and challenges arise at each stage of the research process. This focus lets us highlight (a) the crucial issue of selecting sources and building data; (b)

[1] The bibliometric communities, identified through network analysis, could also be called clusters. But we opted for "communities", a term often used in network analysis, in order to distinguish them from the "semantic clusters".

[2] To be clear, we don't think that quantitative methods are only helpful for very large corpora. However, it is where their surplus value may appear as the most obvious.

how even simple quantitative assessments can be informative; (c) the challenges and subjective choices involved in building and adapting tools; and (d) why interpreting results demands careful attention to various indicators and close knowledge of both the corpus and its historical context. By tracing, step by step, how we assemble and analyze our corpus, we aim to provide a practical example and a reflection on broader methodological issues faced by historians of economics in quantitative inquiries. The analysis of the results relies on the use of an interactive application that we have built for this article.[3]

From sources to corpus

From sources to data

Discussions on quantitative methods often focus on the varieties of existing methods. In practice, however, implementing a method and interpreting its results come last. A great deal of effort goes into collecting, cleaning, and structuring data, much of which remains invisible in published work. Sources rarely arrive as ready-to-use datasets; they must be transformed from somewhat raw materials into usable corpora. In short, the quantitative historian is often a data wrangler as much as a data analyst.

Bibliometric databases are a convenient way to access corpora of economic texts: they record relatively well-structured data, gathering key features of scientific output—authors, journals, affiliations, *etc.* Some of these databases, such as those provided by *Web of Science, OpenAlex,* or *Scopus,* record citation data, which enables tracing intellectual influence through references and mapping scholarly contributions over time.

Regarding data availability, each database has its own strengths and weaknesses. While at a very general level Web of Science, OpenAlex, or Scopus have similar coverage (Martín-Martín et al. 2021; Culbert et al. 2025), some studies might suffer from choosing a less appropriate database regarding the research questions. For instance, Scopus has a lower coverage of the “top five” economics journals before the 1990s and, while being openaccess, OpenAlex has been less curated than the products of for-profit publishers. Whatever the provider, less central or now-defunct journals are more likely to have incomplete or discontinuous digital coverage, impeding their inclusion for historical analyses. Additionally, as these databases are oriented toward English-speaking contributions, they may be inadequate for a research project targeting other languages.[4] More generally, citation practices have only standardised progressively in the postwar period. Consequently, citation data are most of the time relatively poor before the 1960s. Last, although they provide useful metadata and citation information, these databases generally lack full text, which is subject to copyright and therefore, a large and representative corpus of economics articles cannot be obtained from a single publisher.

Even when citations and full texts are available, the quality of information that can be reliably encoded remains limited. Citation data are difficult to structure, because both the very notion of what constitutes a reference and the conventions governing its recording have changed over time—for instance, from references embedded in footnotes to the development of standardized bibliographies and the author–date system (Grafton 1999). Full texts face similar limitations: mathematical expressions and empirical material, such as tables, are often poorly captured by providers and difficult to encode properly. These shortcomings restrict what can be studied by quantitative tools.

[3] The application is available here: https://019adac8-81d4-aa0e-808c-08861c261fd2.share.connect.posit.cloud/

[4] One of the main challenges for a quantitative history of economics is to move beyond reliance on proprietary digital libraries and to actively develop open, historically inclusive corpora. This includes incorporating materials produced in non-Anglophone countries and expanding the range of textual formats considered—such as books, book reviews, working papers, and conference proceedings.

In short, the availability and quality of data determine what can be asked and so answered. The scarcity and structure of data affect every stage of inquiry, from the choice of research questions to the interpretations of quantitative results. Scholars are not condemned to rely on existing databases and may build handmade textual and citation datasets from scratch. However, such tasks remain time-consuming beyond small-scale study. More commonly, it is often necessary to combine information from different databases to fulfill a specific goal. For example, a medium-scale study of the publications of the European Economic Review (Goutsmedt and Truc 2023)—few thousands documents—required to combine three heterogenous databases: Econlit (for JEL codes classification) Web of Science, and Scopus (due to incomplete coverage of the EER in Web of Science).

For our study of rationality, the first step was to identify the best sources. For citation data, the Web of Science (WoS) provides areliable and consistent coverage for our period and has been widely used in the history of economics (Culbert et al. 2025; **martin-martinGoogle2021?**; Claveau and Gingras 2016) . As for full text, we relied on three providers. First, JSTOR's full-text collection offers good-quality scans of most leading economics journals (Jstor 2025). Second, we used Scopus to identify peer-reviewed economics journals not included in JSTOR and to compile a complementary list of articles; for these, we retrieved full texts first through the Elsevier Full-Text API, when available.[5] Third, remaining full texts were obtained through the ISTEX project (Istex 2025), which provides access to a substantial corpus of documents for researchers affiliated with French universities.[6] We obtain a first meta-corpus of 290 000 economics articles that will be later filtered to focus on "rationality".

The Figure 3 shows the distribution of this meta-corpus across the years and languages. Our corpus disproportionately represents Anglo-Saxon journals, which happen to be the most systematically digitized and preserved. This skew is problematic on two fronts. First, it risks marginalizing research traditions poorly represented in English-language articles. Second, it introduces a form of presentism: the retrospective accessibility of these journals' data today should not be conflated with their historical centrality, nor should it imply that journal articles constituted the dominant medium for the circulation of scholarly ideas. Their prominence within the meta-corpus reflects less their past influence than their greater capacity, through resources and institutional support, to maintain comprehensive digital archives. Conversely, the materials missing "are unlikely to be missing at random" (**stoltzMapping2024?**).[7]

Once access to full text is secured, another crucial step is to transform them into a usable database. While WoS citations are already delivered in a relatively structured form, full texts require substantial processing. In most cases, data are not given but result from a process that involves cleaning, categorizing, and selectively removing or reformatting information from raw sources in order to produce a dataset suitable for computational analysis. This challenge is particularly true in the history of science and ideas, and more specifically in the history of economics, where the primary sources often are the texts themselves. These texts contain layers of content, such as section headings, footnotes, or bibliographic references.

This data wrangling is not a tedious prelude to "real" historical analysis. Manipulating,

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[5] The Elsevier API allows users to retrieve the full text of Elsevier journals such as the *European Economic Review*, provided that they have institutional access to the relevant issues.

[6] ISTEX allows us to retrieve economics journals not available through JSTOR or Elsevier API, such as the *Journal of Economic Theory* and to access specific issues of Elsevier journals for which we lack institutional access (e.g., the *European Economic review* before 1997). For more detailed information on the collection of data, see the appendix.

[7] Bibliometric and textual data themselves have a history; they are "made by institutions, through human processes" (Guldi 2023, 34). Whether digital or material, archives are shaped by "distortions" and "occlusions", and the processes through which they are constituted are themselves worthy subjects of historical investigation (**dastonSciences2012?**).

cleaning, and structuring raw sources is a fundamental and often generative stage of research. As Lemercier and Zalc (2019, 62) reminds us, collecting and categorizing sources is also "a moment to reflect on the sources and the purpose of the research." Direct engagement with raw data can prompt the reevaluation of initial hypotheses and the emergence of new questions. In our case, preparing full-text documents raises choices about what counts as a relevant economic text. Should we include book reviews, working papers, or conference proceedings? How should we define an "economics journal"—restrict it to core outlets or extend it to interdisciplinary venues where economics appears regularly? Even within a single article, boundaries are ambiguous: should abstracts, footnotes, or appendices be analyzed or excluded? Each decision may carry historiographical implications. It shapes the corpus and, ultimately, the history of rationality our methods can reveal. There are rarely definitive and uncontestable choices, but rather a series of trade-offs that should be made explicit and required to engage with some of the material available.[8]

In light of this article's purpose, we operated a series of choices in extracting textual data from the full-text materials provided by JSTOR, Elsevier, and ISTEX. Our first choice was to restrict the analysis to English-language articles, since cross-language comparisons involve additional challenges, which would go far beyond the scope of this article.[9] We also restrain our analysis to research articles and filter out book reviews, comments, editorial reports or obituaries.[10] While such materials can illuminate how rationality was debated, they are not primary sites for articulating new ideas within the discipline. Moreover, textual data are by nature voluminous, and filtering improves tractability for large-scale computation. At the document level, we focused on the body text and removed (as far as possible) peripheral elements such as acknowledgements, references, or appendices. We also focus on natural-language text and remove other information that is poorly OCR-processed and often unusable, such as mathematical formulas and data tables.

To illustrate how corpus delineation and cleaning constitute an iterative process, it was only after an initial round of exploratory analysis that we noticed the absence of important economics journals for our purpose in JSTOR. While preliminary results clearly indicated that the rise of behavioral economics had impacted how economists discussed rationality, our domain expertise suggested that certain journals like the *Journal of Economic Behavior & Organization* or the *Journal of Behavioral Economics*—later to be the *Journal of Socio-Economics* and then the *Journal of Behavioral and Experimental Economics*—were missing precisely where they should have figured prominently. This prompted us to further explore potential biases in JSTOR and to augment our corpus with new full texts extracted from Elsevier and the ISTEX project. Exploring raw sources is thus an important step that requires to already engage with both collected materials and the existing literature.

The use of several databases also raised specific issues. Here, our goal was to combine our textual data (from the three providers mentioned above) with citation data from WoS. Achieving this required linking documents across datasets, yet WoS does not provide DOIs, which would otherwise serve as convenient unique identifiers. It therefore fell to us to develop matching procedures to determine whether an article retrieved from JSTOR or Scopus corresponded to the same article indexed in WoS. However, matching based solely on the title is unreliable. For example, the title *"Inflation and Unemployment"* may refer either to James Tobin's (1972) AEA presidential lecture or to Milton Friedman's (1977) Nobel lecture.

[8] The programming dimension of such computational approaches enable iterative feedback between analysis and corpus cleaning (**nelsonComputational2020?**); critical scrutiny can therefore be continuously exercised at each stage of the research "pipeline" (**leeRole2020?**).

[9] Indeed, most textual methods are designed to identify commonalities within texts that share a specific linguistic structure. Hence, the same topic discussed by two documents in two different languages will likely not be identified as related because linguistic differences mask underlying semantic similarity.

[10] This filtering draws on JSTOR's and Scopus' own classifications, supplemented by some additional filtering from ourselves. Despite these safeguards, a perfectly clean restriction to research articles was not feasible, and some residual "non-article" items likely remain.

Consequently, we had to supplement title information with additional metadata, including the journal name—which required standardizing journal titles across databases—as well as the publication year, volume, issue, and page range. Differences in journal coverage and in data storage practices (particularly title formatting) mean that it is generally impossible to achieve a complete match between databases. In our case, approximately one-quarter of the full-text documents after 1945 could not be matched to WoS records, which prevents us from analyzing their citation data.[11]

From data to corpus

In parallel with the transformation of sources into data, we also needed to establish the boundaries of our corpus, by selecting materials either *ex ante*, when choosing which sources to include, or *ex post*, when filtering the collected data. In the context of our project on the history of rationality in economics, we had to determine what qualifies as "economics," and second to identify which documents or parts of documents can be considered "texts" about rationality.

The first challenge was thus to define the extent of our corpus a priori, i.e. to delineate economics as an object of study. Some research objects are relatively easier to delineate, and the transition from a database to a well-defined corpus is therefore straightforward. For example, writing the history of a particular journal (Edwards 2020; Charles et al. 2025) or of one or several individuals (Truc 2025; **andradaUnderstanding2017?**; Fontana and Iori 2023) entails comparatively definitional or boundary challenges. While some large-scale studies focus on the discipline as a whole (Claveau and Gingras 2016; Ambrosino et al. 2018; **bacciniExploring2025?**), such work still requires an operational definition of what are documents in "economics." This definitional issue becomes even more pronounced when the object of study is a specific "field" or "research specialty" [(**morrisMapping2008?**); (**lyutovMachine2021?**);] .

Non-quantitative historical approaches often address large objects by focusing on clearly identifiable cases or canonical contributions, allowing the scope of inquiry to remain flexible and progressively defined. For instance, a historical inquiry on rational expectations may take Muth (1961) as a starting point, or focus on the reactions to Lucas' (1972) model. Quantitative approaches, by contrast, require the explicit delineation of a corpus from the outset, which entails making operational choices about inclusion criteria. How should a corpus on rational expectations be constructed? Which journals should be included? Should one restrict attention to articles explicitly mentioning "rational expectations," or also consider related formulations such as "model-consistent expectations"? Defining the relevant historical materials therefore involves adopting explicit conventions to approximate the object under study (Desrosières 2011). The act of delimiting a corpus is a convention and such act must be assessed instrumentally, not as a definitive delineation, but as an operational hypothesis tailored to a specific research question or even an act of "drawing impossible boundaries" (**lietzDrawing2020?**; **zittBibliometric2019?**).

Many "proxies" have been used in the history and philosophy of economics to define disciplines and sub-disciplines. For instance, (Fontana and Iori 2023) restrict their economics corpus to allegedly most influential economics journals, the "Blue Ribbon Eight"; (Goutsmedt and Truc 2023) used the JEL codes to select macroeconomic documents; while (Truc 2022) and (Jullien and Truc 2024) relied respectively on citations data and institutional affiliations to identify the boundaries of behavioral economics. However, such approximations have implications for the object under study. Both JEL codes and journal classifications—like JSTOR or Scopus classifications—structure a corpus in ways that reflect their own histories, and researchers must consider how these classifications shape the

[11] See the appendix for more information on the matching process.

boundaries of their material. For example, while the JEL codes for neuroeconomics emerged around the same time as the first publications in the field, the JEL codes for behavioral economics appeared more than two decades after the earliest contributions (Truc 2023). A thorough knowledge of the history of the object studied is thus crucial for evaluating the adequacy and representativeness of the resulting corpus.

In our case, we define *economic* documents by building the corpus from *peer-review* journals. This convention has both advantages and limitations. On the one hand, it provides a clear and reproducible criterion, relying on a predetermined set of journals, rather than requiring case-by-case decisions about whether individual documents qualify as "economics". Journals also constitute a central institutional marker of a discipline, alongside training programs and professional associations. On the other hand, this choice excludes other publication outlets, such as books and working papers.[12] In addition, it only partially captures interdisciplinarity, missing interdisciplinary journals or economists publishing in other disciplinary journals. Finally, this choice also entails boundary decisions regarding which journals to include, particularly for those at the margins of "economics", due to their ambiguous editorial standards. We eventually settled on a carefully curated list of 329 journals identified as economics journals in JSTOR and Scopus, but excluding journals that are not entirely academic, insufficiently focused on economics, or only founded within the last twenty years.[13] We were able to retrieve 289538 articles published in these journals from 1900 to 2009. These articles form what we call our "meta-corpus".

The second challenge was to restrain our corpus to documents engaging with the issue of rationality. One of the most straightforward proxies are keywords (Truc 2023). Based on a predefined list of target terms (called a "dictionary"), this approach restricts a corpus to documents that mention these terms with at least a given frequency. In addition to its relative simplicity, it works well for specific and unambiguous terms, like "stagflation" (Goutsmedt 2021) or "Agent-Based Models" (Baccini et al. 2025). But this approach may exhibit several limitations in other cases. Indeed, it focuses on identifying instances of a term rather than the concept as a whole. Just think about the various ways rationality could be discussed in the history of economics: beyond the term "rationality", economists employ various expressions that refer to close ideas such as "maximising profits", "expected utility", the "*homo oeconomicus"*, or the"transitivity and completeness of preferences".

Going beyond searching for occurrences of "rationality" and "rational," we could have constructed an extensive dictionary of terms associated with the concept of rationality. It remains difficult to prevent the dictionary-building process from introducing biases—for instance, by omitting concepts that are historically relevant but salient only during specific periods (such as "hedonism"), or by creating a disproportionate dictionary, with many terms related, for instance, to decision theory but few pertaining to macroeconomics or public economics. Consequently, using a list of words constrains the potential for discovery: by setting the boundaries of the dictionary in advance, researchers may unintentionally exclude terms or themes of which they were unaware, and thus remain unaware of them throughout the analysis (**huistraPhrasing2016?**; **underwoodDistant2019?**).

To overcome this issue, we use a Large Language Model (LLM) to identify documents—and in particular specific sentences within documents—that deal with rationality in our corpus of economic articles.[14] LLMs are trained on extremely large collections of text to learn patterns

[12]Adding these other outlets would pose additional challenges. For both working papers and, above all, books, no large-scale databases provide comprehensive full texts or complete reference lists. In addition, working papers raise the problem of potential double counting, as they may eventually be published in economic journals.

[13]We did not filter journals by language a priori. Indeed, many non-anglo-saxon journals also publish English articles at some more or less frequent occasions. It was thus easier to filter by language a posteriori and at the document level, once articles were collected.

[14]Our goal here is not to provide the reader with extensive details on what these models are and how we use them (see the appendix for additional details and references), but rather to provide general intuitions about

in language. The type of models we use, i.e. bidirectional encoder models such as BERT, learns to predict missing words in a sentence or to determine whether two sentences follow each other. Through this training, the model learns billions of internal parameters that capture regularities in vocabulary, grammar, and meaning. When we input text into such a model, it represents sentences into vectors that summarize their context and meaning. The relative position of each vector in this space reflects semantic proximity: standard distance metrics—such as cosine similarity—can then be used to compare, rank, or hierarchically cluster sentences by meaning.[15] This enables us to compare sentences not by the exact words they use but by their underlying meaning. For example, the same term—such as "model"—may carry different meanings depending on the surrounding context (*fashion model* vs *scientific model)* and a LLM can detect these variations. By converting words and sentences into vectors that reflect their semantic usage, LLMs make it possible to treat ideas and conceptual shifts as measurable objects, thereby opening new possibilities for the quantitative study of economic thought.

We rely on Sentence-BERT (Reimers and Gurevych 2019), an LLM designed specifically to produce sentence embeddings, that is, numerical vectors that represent the meaning of a sentence. The model assigns one vector to each sentence, which allows for straightforward semantic comparison: sentences that express similar ideas end up with vectors that are mathematically close to each other. Using Sentence-BERT, we vectorized more than 61 million sentences published between 1900 and 2009 from our meta-corpus. Starting from a set of *sentences* A including "rationality" and "rational", we can compute their centroid, that is the vector obtained by averaging the embeddings of A. We call such a centroid vector a "representative vector", as it encapsulates the semantic meaning of rational and rationality in our meta-corpus. Then, we could retrieve a set of *sentences* B that are the most similar to this centroid. Thus, from a set of *sentences* A explicitly mentioning the word "rationality," we retrieve a set of *sentences* B—which may never mention the terms but likely discusses a related idea such as profit-maximization (see Ash, Chen, and Naidu 2026 for a similar use).

Computing a single representative vector over 110 years raises serious historical issues, though. Because our project is historical, we must account for how LLMs handle temporally situated language. LLMs are trained on billions of texts, the majority of which are recent, and therefore reflect a presentist, numerical view of language.[16] For example, current models struggle to reproduce earlier writing styles and cannot reliably infer the publication date of a text (Underwood, Nelson, and Wilkens 2025). Sentence-BERT is subject to the same limitations, and the sentence embeddings it produces inevitably inherit this bias. Besides, as illustrated by Figure 3, our corpus is exponentially distributed over time, with most sentences drawn from recent articles and the centroid of our *sentences* A would therefore itself be skewed toward recent language. Consequently, the sentences retrieved in set B, those closest to the centroid of A, would predominantly come from recent periods. In other words, our results would reflect a modern understanding of rationality, at the expense of earlier conceptions of the concept.

To mitigate this presentist bias, we adapted the way sentence similarity is computed across time (see Figure 2 for a visual representation of our approach). Instead of comparing a sentence from, say, 1910 directly to a single representative vector constructed from all sentences in the corpus containing "rationality" or "rational," we construct time-specific representative vectors. For each year, we compute a moving-centroid embedding using a symmetric five-year window, based on all the sentences containing "rationality" or "rational".

the use of LLMs, to understand the building of our corpus.

[15] Henceforth, "similarity" between sentences or documents refers to cosine similarity, defined as the cosine of the angle between two vectors in the embedding space.

[16] Moreover, the texts used to train these models are not primarily academic articles in economics. This means that there may be important gaps between the general language patterns the model has learned and the specific vocabulary, concepts, and writing practices of our "domain" (see e.g. Zhang, Rojcek, and Leippold 2025).

Concretely, for the year 1910, we averaged the vectors of all sentences containing "rationality" or "rational" from 1905 to 1915.[17]

This procedure yields a period-specific reference vector that reflects how these terms were used at that particular moment in time.[18] A sentence from 1910 is therefore evaluated not against a general, and likely contemporary, meaning of rationality, but against its historically situated usage. We call these centroids the "representative vectors." They should be interpreted as operational summaries of the semantic contexts in which "rationality" appears within a given symmetric five-year window, rather than as fixed theoretical definitions of the concept. While the representative vectors are not associated with any real sentences, it is close in the vector space from real sentences. Table 1 reports the 5 sentences closest to the representative vector, ranked by cosine similarity for the years 1910, 1950 and 2000.

Finally, for each year, we select from all the sentences from our meta-corpus the 1% of sentences whose embeddings are closest to the corresponding representative vector.[19] This yields a "sentences-corpus" of 499 157 sentences, which is then used for subsequent textual analysis to identify *semantic clusters* that group similar sentences together.

We also used our representative vectors to extract from our meta-corpus the documents that are closest to the corresponding representative vector. Because each document consists of a set of sentences, we can also give a vectorial representation to the document by computing the centroid of its sentence embeddings. Thus, we select the 10% of articles published after 1960 whose centroid are closest to the corresponding representative vector. Table 2 displays the five closest documents for years 1910, 1950 and 2000 where document vectors are computed as the centroid of all sentence vectors within the document. This "articles-corpus", composed of 25 916 articles, is then used in the bibliometric analysis to identify *bibliographic communities* from 1960 onwards.[20]

In the appendix, Figure 1 summarises this whole process of building our two corpora from our different meta-corpus and Figure 2 summarises the construction of the representative vectors.

Exploring the corpus

Simple exploration

Before returning to our specific corpora on rationality and diving into more advanced LLM-based analyses, we first step back and examine simpler quantitative methods applied to

[17]We initially tested a symmetric two-year window. However, particularly in the early periods, this specification resulted in substantial volatility in average usage of the terms "rational" and "rationality", thereby increasing the number of "noisy" sentences (i.e. sentences not clearly related to rationality or tied to highly specific issues). Conversely, longer windows would reduce the advantages of our historical approach, as they would blur temporal variation too much, especially given the rapid growth of the corpus after the 1960s.

[18]Centroids are widely used in the literature on "semantic change" to represent the average embedding of a temporal slice or a cluster of a corpus (Montanelli and Periti 2024). We adopt the centroid rather than the medoid (i.e. the observed sentences whose embedding is closest to all others in the set), as the centroid smooths over idiosyncratic variation across sentences, whereas the medoid may be disproportionately influenced by a single atypical example. While the medoid is an embedding of a real sentence and is easily interpretable, it is very easy to compute the closest embeddings from the centroid (see Table 1).

[19]The choice of the 1% quantile reflects a trade-off between false positives and false negatives. We observed that a more restrictive threshold (e.g., 0.5%) excludes many sentences that obviously deal with rationality in economics, whereas a more permissive threshold (e.g. 2%) includes sentences that are too distant from our core focus, particularly in earlier periods. The 1% threshold therefore appeared as a reasonable compromise. Moreover, a moderate number of false positives does not substantially threaten the analysis, since such sentences are likely to be classified as "noise" by the HDBSCAN algorithm.

[20]The "articles-corpus" only starts in 1960, because the bibliographic data from WoS are too limited prior to this date to be reliably used.

our broader meta-corpus. Indeed, quantitative approaches come in many forms and levels of complexity. Quantification does not need to be sophisticated to be useful. Simple indicators may not always provide the clearest answers to research questions, but they play an important exploratory role: helping researchers refine their questions, identify anomalies, or detect unexpected patterns.

For textual data, term frequency, that is counting how often particular words or expressions appear, is a straightforward indicator. This metric has been used repeatedly in the literature, especially to track the emergence or decline of fields within economics. In well-delimited domains, term frequency can serve as a reliable proxy for intellectual dynamics. For instance, Truc (2023) shows that counting occurrences of highly specific neuroeconomics terms in economics journals—such as "striatum" or "prefrontal"—closely approximates more advanced quantitative measures, and thus provides a simple but meaningful signal of activity in the field.

Figure 4 shows the relative frequency (with respect to the total number of words published each year) of "rational" and "rationality" in our corpus. The figure reveals a steady postwar increase of the use of both terms, likely reflecting the progressive consolidation of rational choice theory in economics. We observe a marked surge after the 1970s, with a pronounced peak for "rational" in the 1980s, and a smaller peak for "rationality" in the 1990s. The rise of "rational expectations" in macroeconomics and the emergence of behavioral economics following the publication of Kahneman and Tversky (1979) likely contributed to this pattern. Of course, one can only speculate about the precise drivers of these trends from such simple metrics.

While our usage of a LLM will shed light on the context surrounding this upward movement, such context can also be approximated with simple measures. Co-occurrence analysis helps recover the different intellectual settings in which rationality is invoked. Figure 5 reports, by decade, the five words most frequently adjacent to "rational" or "rationality," showing how these associations shift over time. Before the 1930s, the picture was heterogeneous. The notion was tied to the marginalist idea of "calculation," but it referred not only to individual behavior but also to "systems" or "organizations." The discussion was explicitly methodological, as indicated by the prominence of terms such as "method," "explanation," "law," and "foundation." From the 1940s onward, the rise of choice theory places rationality at the center of economic modeling as a device for describing and formalizing behavior. This is mostly visible with the rise of multiple common bi-grams (i.e., combination of two words) that remain stable from the 1930s through the 1980s such as "economic rationality", "rational choice", "rational behavior". By the 1970s—and especially the 1980s—the framework was both extended and contested. First, the most common bi-gram by far becomes "rational expectations" signalling the emergence of a new predominant concept extending rationality. In the 1980s, "rational expectations" appeared 15 more times than the other most common bi-grams. Second, while the concept "bounded rationality" was developed by Herbert Simon initially in the 1950s, the concept only became prevalent during the 1990s with the bi-gram becoming the fourth most common one.

Beyond textual data, citation data also constitute a useful source of information for corpus exploration. The evolution of an idea depends not only on how it is formulated by its authors, but also on how it is appropriated, and reinterpreted by readers. A substantial literature in citation theory examines how citations function as a scientific practice and how they should be interpreted (see Tahamtan and Bornmann 2018 for a litterature review on the issue). Citations may reflect a wide range of motivations—from a genuine desire to acknowledge intellectual debt to more strategic uses aimed at persuading readers or satisfying referees. Nevertheless, citations at the very least signal a relationship that can help trace the lineage and diffusion of ideas, even when the reasons behind the citation are not purely intellectual. Beyond tracing diffusion, highly cited papers tend to be more visible and attract greater engagement—a dynamic that reflects the well-known Matthew effect (Merton 1968). Recent work also shows that citations influence reading behavior and perceptions of quality: highly cited papers are more likely to be read thoroughly and treated as significant intellectual contributions (**teixeiraMatthew2021?**; Teplitskiy et al. 2022; **eikaStarstruck2022?**).

Historians routinely discuss scientific influence and recognition using proxies such as major grants, prizes, and honors. For example, Sent's (2004) narrative of the transition from the dominance of rational choice, through the limited success of "old" behavioral economics (with contributions by, e.g., Simon and George Katona), to the emergence of "new" behavioral economics is organized around such milestones. In this sense, citations serve as a complementary proxy alongside these traditional indicators. For instance, although both Simon and Daniel Kahneman received the Nobel Prize, Kahneman ultimately exerted a broader influence on the discipline—something that is reflected in citation patterns.

Offer and Soderberg (2016) distinguished several citation trajectories among Nobel laureates in economics: those whose citations peak around the award before declining; "innovators with staying power"; "still rising" winners honored before their citation peak; and late winners recognized long after their peak. Figure 6 illustrates these dynamics by plotting citation patterns—both across all Web of Science (WoS) economics journals and within the top five journals—for four seminal works critiquing the standard conception of rationality in economics. Simon (1955) and Allais (1953) represent early critiques of neoclassical rational choice, while Akerlof (1970) and Kahneman and Tversky (1979) are foundational contributions to what economists now refer to as "new" behavioral economics. The latter tradition exerted substantially greater influence than the earlier one, both across all WoS economics journals and within the top five journals.The contrast concerns not only the magnitude of influence, but also the timing and speed of diffusion. Citations to the two "new" behavioral economics papers increased rapidly and steadily from publication onward, whereas Simon and Allais reached only modest citation peaks around the time of their Nobel Prizes—or even later, in the 2000s, amid renewed interest sparked by "new" behavioral economics. As Offer and Soderberg (2016) argue, many laureates benefit from a "Nobel premium," a modest increase in citations following the award. Simon and Allais fit this pattern. Kahneman and Tversky, however, constitute an exceptional case: after the Nobel Prize, their previously declining citation trajectory reversed and rose sharply throughout the period under study.

More advanced exploration

Citation counts are a blunt tool and cannot answer many questions of interest to historians of economics: Who cites these works? Are they cited together? In what intellectual contexts do these citations occur? Moreover, focusing on a small set of references implies a degree of arbitrariness into the analysis. Likewise, simply counting the words that appear next to "rationality" tells us little about whether these words and expressions are used jointly within the same argument or whether they belong to distinct topics and contexts that mobilize the concept differently. It also overlooks the much broader vocabulary related to rationality (e.g., profit maximization, expected utility, social choice).

Most recent quantitative studies in the history of economics rely on what can be grouped under the label of "unsupervised methods." These approaches apply algorithms to a corpus—whether textual or citation data—to generate classifications and assign categories that are not predetermined by the researcher but that emerge from statistical patterns in the data itself, which gives the approach its "unsupervised" character.[21] Assigning such

[21]"Unsupervised" does not mean that the modeller has no influence on the output. The analyst must choose a particular model and its parameters and these choices embed methodological priors about the types of patterns the procedure is likely to reveal. For instance, one crucial parameter in many unsupervised models is the number of categories that will emerge from the algorithm. We detail and justify the different parameters chosen for our methods in the appendix.

categories imposes a form of internal organization on large and otherwise unwieldy bodies of material, enabling an accelerated or "distant reading" (Moretti 2013). The typical output of these methods is a map of a discipline or research area, showing how different entities—topics, articles, authors—relate to one another.

In contrast to unsupervised methods, supervised methods aim to estimate predefined relationships or reproduce categories specified in advance. Supervised methods—such as regression models or classification algorithms—require researchers to formulate hypotheses, define variables, or establish categories prior to analysis (**doAugmented2022?**). Unsupervised methods rather seek to detect patterns, groupings, or structures that emerge from the data itself.[22] This "discovery" capacity (Grimmer, Roberts, and Stewart 2022) is especially valuable in large-scale and long-run historical analysis, where the relevant categories may be uncertain, evolving, or difficult to define in advance. In such contexts, imposing predefined classifications risks introducing categories that are historically inadequate or anachronistic. Unsupervised approaches therefore provide a useful way to identify unexpected regularities, semantic groupings, or intellectual communities before moving to interpretation.[23]

In this paper, we use unsupervised methods to analyse both the articles-corpus and the sentences-corpus. The articles-corpus consists of the 10% of articles whose embeddings are most similar to the annual representative vectors. To make sense of this large set of documents, we employ a method that groups together documents mobilizing similar understandings of rationality. We use bibliographic coupling, a method commonly employed in the history of economic thought (see e.g., Claveau and Gingras 2016; Truc 2022; Baccini et al. 2025). In a bibliographic coupling network, documents are represented as nodes, and links between them are weighted according to the number of references they share. The more references two articles have in common, the stronger their connection—and, in network visualisations, the closer they tend to appear to one another. The underlying premise is that shared references provide a proxy for intellectual proximity. Such networks make it possible to identify communities of closely related documents and to delineate the core-periphery structure of a discipline or research field.

However, citation practices tend to favor more recent works, which makes it unhelpful for historians to construct a single citation network spanning an entire century. Following a method that has proven effective in the field (**claveauMacrodyanmics2016?**; Goutsmedt and Truc 2023; Camilotto 2023), we therefore split our corpus into overlapping eight-year windows,[24] starting in 1960 (1960–1967; 1961–1968; … ; 2002–2009), and built a series of 43 separate networks. Using community-detection algorithms (Traag, Waltman, and Eck 2019), we identify groups of documents that share a substantial fraction of references and therefore have a similar intellectual background; we refer to these groups as "bibliometric communities." The next step is to identify communities that persist over time. When two communities from consecutive networks share many of the same nodes (i.e. articles), we treat them as instances of the same underlying group—an "intertemporal bibliometric community." In this way, our bibliometric communities bring together documents from different periods of time but that are likely to engage with rationality in similar ways, based on their shared citation patterns. This method allows us both to *zoom in* on specific communities at a given period and to

[22] See (**simonsLarge2026?**) for a discussion of using LLMs with unsupervised or supervised methods in the history and philosophy of science.

[23] That does not mean, however, that an unsupervised algorithm can be applied blindly to historical questions: a substantial part of our effort has consisted in adapting the method to our historical data, especially to accommodate the growth of the corpus over time.

[24] The eight-year window for bibliometric communities follows established practice in the quantitative history of economics (**claveauMacrodyanmics2016?**), which is generally between 5 and 10 years (**abramoAssessing2011?**). In our case, economics being a social science with longer citation spans of life (**lariviereLongterm2008?**; **aistleitnerCitation2019?**), and our object of study spanning across multiple decades, a longer time window than 5 years is favored while preserving more granularity than a 10-year one.

zoom out by reconstructing the broader picture of a dynamic intellectual field.[25]

For textual analysis, we use the sentences-corpus that gathers the top 1% of sentences that are most similar to our annual representative vectors of sentences with "rationality" and "rational" (see above and Figure 2). To analyse this corpus, as with bibliographic coupling, we want a method that groups together our observations (here the sentences) that employ similar meanings of rationality. We draw on the literature on semantic change that measures the change of meaning of words over time (Kutuzov et al. 2018; Montanelli and Periti 2024; **peritiSystematic2024a?**) and take inspiration from Giulianelli, Tredici, and Fernández (2020). We therefore cluster the sentence embeddings using the HDBSCAN algorithm, a widely used unsupervised method for clustering LLM embeddings.

Again, historical perspective matters. Because the distribution of identified sentences is strongly present-biased (Figure 3), clustering all sentences at once would risk over-representing recent periods. We therefore perform clustering separately for each decade from 1900 onward (merging the first two decades due to fewer sentences).[26] As in our bibliographic approach, we then seek to form larger groups over time, allowing us to "zoom in" and "zoom out" depending on what we are searching for. Using the cosine similarity between clusters across decades, we merge the closest ones into 17 "intertemporal semantic clusters."[27]

With both methods, we thus obtain a list of "intertemporal bibliometric communities" (from 1960 to 2009) and "intertemporal semantic clusters" (from 1900 to 2009). These allow us to identify subgroups within our corpus across multiple dimensions: different subfields of economics (e.g., behavioral economics, macroeconomics), different conceptions of rationality (e.g., bounded rationality, rational expectations), or even, more heuristically, different methodological traditions (e.g., experimental work, econometrics). Both approaches have clear strengths and limitations. Some are largely inherited from their respective sources. In our case, citation data cannot be meaningfully exploited before the 1960s, while our OCR full texts handle formulas and tables poorly, making economic models—an essential component of our story—less visible in the semantic analysis.

Ideally, relying on *both* methods serves complementary purposes. First, comparing the results of different unsupervised computational approaches—such as LLM-based embeddings and bibliometric analysis—acts as a form of robustness check. The systematic comparison of sources and methods is a long-standing principle in historical research, allowing scholars to triangulate information rather than depend on a single perspective. Second, the two methods illuminate different aspects of intellectual history. Full texts reveal the meanings, contexts, and semantic nuances of rationality as economists used the concept in their writing, independently of their institutional proximity or citation habits. Citation data, by contrast, highlight patterns of intellectual influence and diffusion: who cites whom, how ideas travel across fields, and where intellectual boundaries lie. It conveys sometimes a more sociological dimension: scholars tend to cite the work of people publishing in similar journals, going to similar conferences, etc. The following sub-section illustrates how we interpret our results, and how the two methods complementarity may be useful.

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[25] Refer to the appendix for details.

[26] We don't use overlapping windows in this case, notably for computational reasons: finding HDSBCAN clusters on a set of tens of thousands of sentences is much more computationally intensive than finding bibliometric communities for at most five thousands of articles.

[27] Intertemporal semantic clusters are thus composed of several intra-decade clusters that have been aggregated together. The interactive application represents all these intra-decade clusters and their integration into intertemporal clusters. Refer to the appendix for details.

Interpreting "Results"

Taken together, the textual and bibliometric approaches allow us to study both the semantic and thematic contexts of the uses of rationality in economics and the channels through which these ideas circulated within the discipline. But how should such a "study" proceed? Unlike simpler quantitative tools, unsupervised methods do not produce ready-made results. In our case, they generate statistical categories—bibliometric communities and semantic clusters—that lack predefined conceptual "labels." This strategy—reducing a large and complex corpus to a smaller number of coherent groups—is well established. Yet the groups themselves are created solely on the basis of statistical similarities, and their historical or conceptual significance emerges only through subsequent "exploration" (**simonsLarge2026?**) and "validation" (**grimmerText2013?**). To make sense of what brings texts together, we therefore need indicators that identify shared features and help hierarchise the documents to be read first. At the same time, a good knowledge of the corpus and of the relevant intellectual debates is indispensable for making sense of the raw computational results, even once they have been informed by a set of indicators.

Therefore, whether at the stage of data construction or in the interpretation of statistical patterns, quantitative methods demand a continuous dialogue with qualitative interpretation, grounded in a close reading of the relevant secondary literature. Only through this iterative back-and-forth can we turn algorithmic groupings into meaningful historical insights. This is both a weakness and a strength. On the one hand, the analysis is not immediately transparent, as it requires the construction of intermediate tools—such as our interactive application—to guide interpretation. On the other hand, it is consistent with the practices of historians of economic thought, who likewise select, prioritise, and read texts in order to make sense of their corpus.

For the purpose of the exploration and interpretation of our results, we have built an online interactive application, where one finds all the semantic clusters and bibliometric communities, and their corresponding indicators. Let us take as an example the semantic cluster "Rationality and its Limits," which gathers close to 20,000 sentences from 1920 to 2009.[28] How can we get a sense of what this cluster actually coalesces around? Before any qualitative interpretation or contextualization with the secondary literature, we must first produce a series of indicators to characterize the cluster. Such indicators include:

- The most identifying words of the cluster for each decade, based on Term Frequency-Inverse Document Frequency (TF-IDF). While the first period of the cluster (the 1920s) deals notably with "rationalisation" and "human reason," the 1950s are more directly focused on "rationalism," "rationality," and "rational behavior." After the 1970s and until 2009, "bounded rationality" emerges as a core concept in this cluster.
- The sentences within the semantic cluster that are closest to its year-representative vector as well as the closest sentences from the cluster's centroid itself for each period. The first category tends to highlight how the concept of rationality is discussed, while the second category centers more directly on the core semantic content of the cluster, often offering clues about the topics and debates that animate it in different periods. For instance, in the 1950s, one of the sentences closest to the representative vectors is Simon's claim that "If man, according to this interpretation, makes decisions and choices that have some appearance of rationality, rationality in real life must involve something simpler than maximization of utility or profit" (Simon 1959, 259–60). As for the closest sentences from cluster's centroid, Allan Drazen (Drazen 1980, 294) warned in the 1980s that "Deciding on the suitable notion or definition of rationality is not an easy task," in his survey of the disequilibrium-theory literature in macroeconomics.

[28] One step in the discovery and interpretive process is often the assignment of "labels" to clusters in order to facilitate the interpretation of visualizations.

- The articles with the most sentences in the cluster for each decade. Here we find, for example, Terence Hutchison's "Expectation and Rational Conduct" (Hutchison 1937); Jacob Marschak's "Rational Behavior, Uncertain Prospects, and Measurable Utility" (Marschak 1950); Herbert Simon's 1978 Richard T. Ely Lecture at the AEA meetings (Simon 1978) and his Nobel lecture (Simon 1979); as well as Sugden's 1991 *Economic Journal* survey on rational choice (Sugden 1991). We also observe that many of these articles are published in behavioral-economics journals such as the *Journal of Socio-Economics* and the *Journal of Economic Behavior and Organization*. From the representative sentences and articles, we can also infer recurring authors.
- From the 1950s onward, the most cited references (based on the number of citations per article, whether the article has one or ten sentences in the cluster). In the 1950s, one of the most important references within this cluster is John von Neumann and Oskar Morgenstern's *Theory of Games and Economic Behavior* (Von Neumann and Morgenstern 1944), while Kahneman and Tversky (1979) becomes the most cited reference in the 2000s.

With all these indicators—and prior to any further qualitative investigation—we can already identify this semantic cluster as centrally relevant to our study, since it bears directly on the concept of rationality itself (hence our label, "Rationality and Its Limits"). Our textual approach is therefore capable of isolating specific uses of rationality—here, debates concerning the meaning of rationality itself and the limitations of its application in economics—thereby enabling us to trace the evolution of these uses over time as well as across topics and subdisciplines, including game theory, macroeconomics, behavioral economics, and evolutionary economics.

The bibliometric approach does not allow for such temporal or cross-subdisciplinary exploration. In our bibliometric communities, no grouping addresses rationality and its limits in general across multiple decades. Rather, bibliometric analysis identifies groups of articles from authors in close academic communities. In relation to the semantic cluster "Rationality and its Limits", we observe, for instance, the emergence of the community "Behavioral Economics: Risk & Uncertainty" community in the 1981-1988 window. As with the semantic clusters, for each bibliometric community in each temporal window we extract, notably, the sentences closest to the representative vectors—based on the articles belonging to the community—as well as the most cited references and the most identifying words. We also examine how each community in a given window originates (i.e., whether it derives primarily from the same or from different communities in preceding periods) and what it becomes in the following period (i.e., its subsequent "destiny").

Discussion

Equipped with these groupings and indicators, we can make sense of our results. How can they help enrich, complete, and refine our understanding of the various and evolving meanings of rationality in economics? This paper does not propose an alternative history of the concept, which would extend far beyond the scope of this methodological discussion. Rather, we highlight selected findings that corroborate and strengthen strands of the existing literature, notably by allowing us to make claims about the prevalence and timing of considerations about rationality, while also broadening the scope of that literature and opening avenues for further research.

Throughout much of the period—but especially before the 1940s—sentences captured by our representative vectors refer sometimes less to the rationality of economic agents than to the rational character of economists' arguments, understood in terms of coherence or sound reasoning. This pattern reflects the logic of our semantic approach: LLMs tend to associate explicit mentions of rationality with broader discussions of reasoning and coherence, even when the connection to later, behavior-centered notions of rationality remains indirect. We retain this ambivalence deliberately since it reveals both historical semantic change and the interpretative challenges inherent to large-scale analysis.

Our goal is also to show how we navigate the results and triangulate indicators to produce coherent historical narratives. While we rely on the secondary literature, the articles we foreground are those identified by our indicators as some of the most significant for understanding the semantic clusters or bibliometric communities.[29] We provide two complementary discussions: a macro-level perspective that highlights broad patterns in the evolution of rationality, and a micro-level analysis that focuses on the emergence of a specific topic—behavioral economics.

Through the Telescope: Broad Patterns on the History of Rationality

When our quantitative inquiry begins in 1900, the concept of rationality had already gained prominence, in line with the rise of scientific and abstract reasoning in the late nineteenth century. The scope of rational behavior was closely tied to debates on the proper domain of economic analysis (Giocoli 2003). With the marginalists, the economic agent was increasingly defined by calculative capacity. Jevons, for instance, portrayed the agent as a "calculating machine" balancing pleasures and pains, a view later formalized as utility maximization (Maas 1999).

In our first two decades, semantic cluster "Utility Theory" represents discussion centred on utility and brings together insights on marginal utility from, among others, Sidney. J. Chapman, Arthur Pigou, Francis Y. Edgeworth, and John M. Clark. Cluster "History of Economic Thought & Philosophy of Economics" also captures discussions about utility: arguments about utility were framed as interpretations, critiques, or extensions of classical authors—most notably Mill, Ricardo or Malthus. Yet, if anything, our results suggest that marginal utility theory appears mostly under severe criticism in this period, especially from institutionalist economists such as Thorstein Veblen, Wesley Mitchell, John Maurice Clark, and Rexford Tugwell.[30]

The criticism was directed along two main lines. First, as Giocoli (2003, 48–50) points out, institutionalist economists rejected the "unfounded psychological underpinnings of neoclassical economics" and sought to replace them with sounder foundations (see also Yonay 2001, 101–6). They proposed instead to renew the economic method by drawing on psychology and its emerging "behaviorist" approach (Rutherford 2001, 175–76). Semantic cluster "Methodological Discussions" captures a significant share of sentences from 1900–1920 that articulate such institutionalist critiques. Institutionalists were energized by new developments in psychology. In his review of the literature on "Human Behavior and Economics," Mitchell emphasized the growing interest of economists in psychological matters, which stemmed from "a somewhat tardy recognition that hedonism is unsound psychology, and that the economics of both Ricardo and Jevons originally rested on hedonistic preconceptions" (Mitchell 1914, 1). For economics, the crucial issue was thus to choose "between providing a sounder psychological basis for our analysis, and holding that its

[29] Of course, as in any historical inquiry of this kind, we are not entirely protected from selective emphasis, all the more so given that the scope of our study is too broad to summarize every pattern revealed by our results. Other scholars could have, at various junctures, pursued different interpretive directions on the basis of the same indicators (especially most similar sentences and articles). The publication of the interactive application we used to explore the results thus serves as an exercise in transparency and—in addition to being, we hope, useful for historians of economic thought more generally—allows readers to assess the robustness and limits of our narrative.

[30] However, because we focus on English, published articles, our clusters are biased toward American journals, notably in the first decades.

psychological basis does not concern the economist" (2).[31] Such criticism did not falter in the 1920s. In the cluster "Rationality and its Limits"—centered at the time on "rationalisation" and "human reason"—Tugwell (1922, 317) encouraged economists to move beyond the "rigid classical *homo economicus*" and noted that "psychologists have already pretty well revolutionized the scientific definition of human nature."

Second, for institutionalist economists, institutions were indispensable for understanding economic behavior, and different institutions implied different behaviors; there was no single, context-independent *homo oeconomicus*. In the cluster "Methodological Discussions", in his review of John Bates Clark's contribution, (**veblenProfessor1908?**) argued against the "statical scope and character" of marginal-utility theory, which, in his view, cannot address "questions of genesis, growth, variation, process." Similarly, in the cluster "Utility Theory", E. H. Downey (1910, 253) claimed that "Marginal-utility economics has nothing to say of the genesis, growth, or current working of economic institutions" (see also Rutherford 2013, 141). Its "deliverances," he argued, were therefore "futile for the problem of social betterment," since such "betterment" concerned the adjustment of institutions to changing circumstances and ideals. In the early twentieth century, economic behavior, according to institutionalists, had to be understood through the prism of the "pecuniary institutions" that shaped it. For instance, in the cluster "Methodological Discussions", Charles Cooley (Cooley 1915) underlined that "pecuniary valuation," or economic valuation, is only possible because of the institutions that make such valuation possible (see also Rutherford 2013, 58–59). Figure 5 shows well the importance of "pecuniary" as a neighbor to "rational" and "rationality" in the first twenty years of the twentieth century.

The centrality of institutionalist economists—as well as the prominence of U.S. journals in our corpus—is visible in the scope of the semantic clusters in the first periods: sentences related to rationality, in addition to the clusters already mentioned, were grouped into themes related to the railway industry, progressive taxation (see cluster "Legislation, Regulation & Planning"), tariffs and commercial policy in the United States (see cluster "Price Theory"), or agriculture (see cluster "Agricultural Economics"). In line with institutionalism, most of these economists examined these themes through the laws and rules governing economic activity, focusing on how institutions structure the behaviour and interests of collective actors rather than on abstract individual assumptions. Within these clusters, "rationality" is rarely invoked explicitly, although discussions of the behavior of "business men" (cluster "Firm Theory & Entrepreneur"), "workmen" (cluster "Legislation, Regulation & Planning") or "owners of land" and "farmers" (cluster "Agricultural Economics") were common. Across successive periods, however, these thematic configurations declined in prominence, and most intertemporal clusters associated with them disappeared after the Second World War. Indeed, the period from 1930 to 1959 displays substantial transformations in the structure of debates concerning rationality.

The most obvious transformations are those described by Giocoli (2003) as "the escape from psychology" and, more generally, the shift from a "system-of-forces"—"economic processes generated by market and non-market forces"—to a "system-of-relations," in which the aim becomes to establish the existence and properties of equilibria through the "validation and mutual consistency of given formal conditions" (Giocoli 2003, 4–5). "Rationality" was progressively distinguished from "reason" or "intelligence" and recast as a formal, axiomatic notion—"rigid rules that determine unique solutions" (Erickson et al. 2013; see also Klaes and Sent 2005).

While intertemporal semantic cluster "Utility Theory" constituted only a tiny share of sentences between 1900 and 1919, and was absent in the 1920s, it gains importance in the

[31] Some years earlier, Mitchell (1910, 97) already pushed forward similar claims, regretting that "few economists have regarded the study of psychology as a necessary part of the equipment of their work," often relying on "tacit preconceptions" rather than to seek to "psychologists to gain a knowledge of the mind and its modes of operation."

1930s to the 1950s (though still representing less than 5% of all sentences). During this period, the heart of the debate concerned the measurability of utility and the opposition between "cardinal" and "ordinal" utility. In the 1930s, the debate still bore on the psychological dimension of utility measurement. In 1933, John Hicks and Roy Allen (1934) coauthored an article on demand analysis that eliminated marginal utility by appealing to the concept of the marginal rate of substitution—the slope of the indifference curve (see Moscati 2019, 98–100). Already in 1932, Allen had made this project explicit: by starting from "preferential discrimination," or individuals' preferences over different bundles of goods, it becomes unnecessary to "make any assumption about the existence of 'total utility' or about the measurability of 'utility'; the hedonistic hypothesis has been rendered superfluous" (Allen 1932, 207).[32] In other words, "Subjective and psychological concepts have been discarded from pure economic theory" (ibid.).

Yet this position was not uncontested. Indeed, in these debates, terms such as "introspection," "satisfaction(s)," and "pleasure" recur frequently in the cluster during the 1930s and 1940s (see TF-IDF, and, for instance, Armstrong 1939). But they disappear from the top identifying words in the 1950s. During this period, debates on ordinal versus cardinal utility continued—with a renewed defense of the cardinalist position and ongoing disputes about the very meaning of "measuring utility" (Moscati 2019, chap. 10)—yet these discussions had largely emancipated themselves from psychological or behaviorist considerations. These debates are also associated with a new intertemporal semantic cluster "Decision Theory", distinct from the cluster "Utility Theory." Appearing in the 1940s and expanding substantially during the 1950s—to nearly 8% of all sentences—this cluster centers on the formalization of agents' preferences and choices. Here, we encounter what would become the standard vocabulary of decision theory: "individual orderings," "preferences," "decisions," "transitive," etc., and Von Neumann and Morgenstern (1944), Friedman and Savage (1948), and Savage (1954) are the most cited references by the core articles of the cluster. To complete this panorama, a separate cluster "Game Theory" emerges in the 1950s, which ultimately comes to represent nearly 15% of all sentences after 1990.

Transformations in the conception of rationality are also visible in intertemporal semantic clusters that persist over time while reflecting profound shifts in the discipline's methodology and the growing centrality of rational choice theory. The cluster "Firm Theory & Entrepreneur" spans the period from 1900 to 1979 and therefore provides a useful vantage point from which to trace how changing conceptions of rationality reshaped the treatment of firms' decisions. In the 1920s, the cluster emphasizes entrepreneurial decision-making, understood primarily as prudence facing uncertainty rather than formal optimization. From the 1940s, this cluster increasingly centers on "profit maximization." The cluster captures the core of the so-called "marginalist controversy" (Backhouse 2009), notably through Richard Lester's attack on marginal theory and Fritz Machlup's (1946) response. Machlup defended profit maximization and, more broadly, marginalist theory itself. He rejected the evidential value of Lester's questionnaire-based surveys of entrepreneurs, while Lester replied that "at the heart of economic theory should be an adequate analysis and understanding of the psychology, policies, and practices of business management in modern industry" (Lester 1947, 146).[33] More generally, the marginalist controversy provided a key intellectual background for Friedman's essay on positive economics (1953), in which he formulated the "as if" principle to justify, among other assumptions, profit maximization. By the 1950s, the most

[32] See closest sentences to the "Utility Theory" cluster's centroid.

[33] Machlup's position was that even if a "goodly portion of all business behavior may be non-rational, thoughtless, [or] blindly repetitive" (1946, 520), questionnaire evidence and empirical findings more generally did not demonstrate the failure of marginal theory when properly interpreted. In a similar defense of profit maximization, Leonid Hurwicz (1946, 110) first acknowledged that it was not "inconceivable that business is run more by routine than by rationality," but argued that behavior may appear irrational or merely "routine" from the standpoint of a purely static theory that ignores uncertainty, while proving fully "rational" once uncertainty and long-run effects are taken into account (see also Herfeld 2018).

cited articles within the cluster "Firm Theory & Entrepreneur" were Machlup's (1946) and Friedman's (Friedman and Savage 1948), both of which relied heavily on "as if" reasoning, notably to justify, in the latter, the behavior of agents regarding expected utility.[34]

Shifts in the discipline's methodology are also captured by the cluster "Methodological Discussion." While in the 1920s and 1930s economists still engaged with notions such as "human nature," "passions," or "prejudices," the postwar period saw the growing prominence of terms such as "logical implications" and "assumptions," followed, from the 1970s onward, by an increasing focus on "models" and "optimality."

It thus becomes clear that by the 1950s the conception of rationality had been profoundly transformed, moving in a more formal direction and largely emancipated from psychology. Another transformation also deserves emphasis: as rationality was recast in terms of consistent choice rather than as a psychological portrait of *homo oeconomicus*, the normative dimension expanded—from individual behavior to questions of social choice and of "rationalizing" policies (Erickson et al. 2013; Hands 2015).

The emergence of this social dimension of rationality is already observable in the interwar period, well before the Cold War. In the 1930s, the concept of "rationalization"—relatively neglected in the history of economic thought—was central in the cluster "Rationality and its Limits." In a literature review on this concept, Robert Brady (Brady 1932, 526) remarked that "scarcely any term… has occasioned more discussion and dispute." The concept referred to "every technique, program, or plan of organization which promised to promote (…) the 'efficiency' of individual enterprises, entire industries, or even the total of economic processes nationally or internationally considered." Attention to "rationalisation" is also visible in the cluster "Agricultural Economics", with the rise of "farm management" aimed at improving farmers' living conditions. It becomes even more pronounced with the multiplication of debates on economic planning and the rationalization of policy decisions from the 1940s onward, particularly in the cluster "Legislation, Regulation & Planning."

As rationality increasingly came to be framed through modern microeconomics, the concept was extended to collective choice. In the intertemporal semantic cluster "Decision Theory", the rational "decision making" is applied not only to individuals but also to firms and public policy. Social choice issues emerge in this cluster in the 1970s, notably through debates on the implications of Arrow's impossibility theorem (1951; see e.g. Igersheim 2019).[35] Arrow's contribution also stimulated the formation of the cluster "Voting & Public Choice" in the 1960s, where it appears as the primary reference, alongside works by Downs (1957) and Buchanan and Tullock (1962). This cluster is closely associated with publications by *Public Choice* (see Cherrier and Fleury 2017). Welfare economics and public choice, as well as public finance (Desmarais-Tremblay, Johnson, and Sturn 2023), are also clearly visible in the bibliometric analysis after 1960. Indeed, one of the major bibliometric communities of the 1960s "Public Economics: Externalities" with James Buchanan, Ronald Coase, and Richard Musgrave as central references. The prominence of public finance increases further in subsequent decades, with many communities gravitating around "Public Economics: Externalities", notably one devoted to optimal taxation after 1968; taken together, these different "Public Economics" communities account for more than one fifth of the entire network in these periods.

We have seen how the meaning of rationality progressively transformed and formalized after the 1930s, and how the rationalization of collective decision-making became central in the postwar period. So far, however, we have said little about macroeconomics. Macro-oriented

[34] The cluster "Firm Theory & Entrepreneur" nevertheless remained a site for critiques of profit maximization. During the 1960s and 1970s, contributions that proposed alternative frameworks for explaining firms' decisions were highly cited (such as Simon 1959) or appeared among closest articles from centroid (like Winter 1971).

[35] This grouping of sentences in the 1970s actually took its origins in the cluster "Utility Theory" (which ended in the 1960s) with a focus on welfare economics.

issues constituted a relatively important semantic cluster from the beginning of our period, in 1900, notably with the intertemporal semantic cluster “Monetary Policy & Rational Expectations). This cluster grew in importance in the 1930s, in connection with debates surrounding Keynes’s work, liquidity preference, and unemployment. It is only in the 1970s, however, that this cluster becomes dominant in our panorama of rationality. This shift is closely linked to the emergence of the concept of”rational expectations,” which profoundly transformed the way rationality was discussed in economics. The concept was developed by Muth (1961) but only later popularized when Carnegie colleagues—Robert E. Lucas, Edward C. Prescott, and Thomas J. Sargent—applied it to macroeconomics (Lucas 1972; Sargent and Wallace 1973). By the late 1970s, it had circulated beyond academia into policy circles and the press, and was often described as a “rational expectations revolution” (Duarte 2025). Our methods help account for this rapid diffusion beyond controversial policy debates during the stagflation era. First, the results show no significant trace of rational expectations in the 1960s in either the bibliometric or textual outputs.[36] But from the 1970s onward, it occupies a central place in the literature on rationality, reaching its apogee in the 1980s, when it accounted for nearly 30% of all sentences in the textual analysis, as well as a comparable share of articles in the bibliometric analysis, spread across several bibliometric communities (“Macroeconomics: Business Cycles & RE,” “International Macroeconomics,” and “Macroeconomics: Modelling Consequences of RE”).

A similar trajectory to macroeconomics can also be observed in finance. The origins of financial issues can be traced to the semantic cluster “Capital & Investment Theory,” which focused initially on the theory of investment. Until the 1930s, the issue was framed around the returns of capital. For instance, the cluster includes the so-called Hayek-Knight controversy on the conception of capital (Cohen 2003). After the 1940s, decision-making under uncertainty becomes a central theme, crystallized by works such as George Shackle (1942) on investment under radical uncertainty, which figures among the most representative articles. One decade later, Jack Hirshleifer (1958) inscribed investment theory within the rising neoclassical conception of rationality, by formalizing investment decisions as a problem of intertemporal maximization of consumption. In the same vein, the Modigliani–Miller theorem (Modigliani and Miller 1958) further consolidated this shift by grounding firm financial decisions as an arbitrage problem, opening the path to modern financial economics. Their theorem, they argued, can be used “as a basis for rational investment decision-making within the firm” (Modigliani and Miller 1958, 296). After the 1960s, the cluster “Capital & Investment Theory” became much more prominent, accounting for more than 10% of all sentences. In this period, rationality is increasingly framed in formal decision-theoretic terms and focused on appropriate investment choices under uncertainty. Hence, our analysis shows that rationality in finance enters primarily through corporate finance, rather than through asset-pricing issues such as market efficiency, where rationality long remains an implicit assumption, rarely discussed explicitly prior to the emergence of behavioral finance.

However, from the 1980s onward, modern asset pricing becomes dominant within the cluster “Capital & Investment Theory” as indicated by the growing prominence of terms such as “assets,” “arbitrage,” and “capital asset pricing”. We also observe increasing proximities between the finance literature and the rational-expectations framework from the mid-1970s onward. This is visible both in the textual and bibliometric analysis, in particular with the “Finance: Market Information” bibliometric community, which emerges in the 1971-1978 window and focuses increasingly on rational expectations. A central contribution found in both semantic and bibliometric clusters is Grossman and Stiglitz (1980), which draws on Lucas’s imperfect-information framework (Lucas 1972) and combines rational expectations with noisy signals to question market efficiency. More generally, rational expectations

[36]This highlights a key caveat of our methods. Because these methods foreground statistically “significant” patterns and indicators’ prevalence, they are not always well suited to tracing and interpreting the origins of a concept or theory with the care that close historical study and archival research provide.

models—initially developed in a macroeconomic context—became a central framework for analyzing the (ir)rational use of information in finance (see Delcey and Sergi 2023).

The 1970s and 1980s thus marked a relative dominance of macroeconomics and finance, with comparatively less weight given to debates on individual rationality. This situation begins to change gradually from the 1980s onward. In the 1981-1988 bibliometric network, the community "Behavioral Economics: Risk & Uncertainty" appears (representing around 5% of the network), capturing the early emergence of behavioral economics as promoted by Kahneman and Tversky. In the 1990s, the rise of behavioral economics—together with new approaches in game theory—fundamentally altered this configuration, leading to more specialized investigations of individual rationality, which now dominate and are distributed across multiple bibliometric communities. This transformation represents not merely a shift in research topics, but a profound reframing of how economics approaches rationality: from an assumption embedded in theories and models to an object of direct investigation at the individual level.

We have sought here to offer a broad panorama of the major transformations in the concept of "rationality" in economics. This approach allows us to identify large-scale changes when they become visible in the data—namely, when particular clusters come to represent a significant share of sentences (or of the bibliometric networks). However, historians of economic thought are often concerned with understanding how such transformations came about: which contributions, authors, and communities played a decisive role. Addressing these questions requires a more focused perspective, tracing the evolution of specific topics and literatures. This is the aim of the next section, which moves beyond the panoramic view to provide a more detailed narrative of the rise of behavioral economics outlined above.

Under the Microscope: Simon's Reception and the Birth of Behavioural Economics

The term "behavioral economics" has a complex history. While it is now mostly associated with the work of Kahneman and Amos Tversky, George Katona is often credited as an early adopter of a behavioral approach (Gilad, Kaish, and Loeb 1984; Hosseini 2011). To distinguish among different strands of behavioral economics, Sent (2004) proposed an influential distinction between "old" and "new" behavioral economics. This distinction separates the heterogeneous and reformist efforts of Katona, Simon, and others to challenge mainstream economics from the more homogeneous heuristics-and-biases research program developed by Kahneman and Tversky. Sent argues that "new" behavioral economics succeeded in part because it was less radical than earlier approaches and because it emerged in the 1980s, at a moment when economic theory faced multiple epistemological challenges, thereby creating an opportune context for alternative frameworks.

The divergent fates of Simon's bounded rationality and Kahneman and Tversky's heuristics-and-biases approach thus present a striking puzzle in the history of economic thought. Both challenged standard assumptions of rationality under the banner of behavioral economics, and both were ultimately recognized with Nobel Prizes, yet their patterns of reception and adoption differed. While our methods cannot identify the precise epistemological mechanisms behind this divergence, they allow us to trace and compare the trajectories of reception and influence of these two approaches, shedding light on how and when they followed different paths.

Our analysis suggests that Simon's concept of "bounded rationality" stands as one of the most influential critiques of standard rationality in economics, particularly in its role in structuring debates about rationality itself. Simon's work dominated discussions in the 1960s and 1970s, with his seminal articles (Simon 1955, 1959) and his Nobel lecture (Simon 1979) appearing across a wide range of research areas, from social choice theory to the theory of the firm and price setting. The textual analysis highlights Simon's conceptual centrality. Semantic cluster "Rationality & its Limits" undergoes a marked transformation following the rise of Simon's influence. In the 1950s, the discourse was centered on terms such as "rational behavior," "rationalization," and "irrationality." By the 1970s, "bounded rationality" had become one the most prevalent expression in the cluster, and by the 1990s, references to "boundedly rational agents" and "procedural rationality" clearly positioned Simon's concepts at the core of economic discussions, reflecting a delayed but substantial integration into the discipline.

Moreover, Simon's impact is visible not only in the changing language of rationality but also in the growing prominence of these debates within economics. The cluster "Rationality & its Limits" expands from 5.76% of all sentences in the 1950s to 7.85% in the 1980s, peaking at 9.56% in the 1990s. This growth signals a significant increase in the attention devoted to foundational questions of rationality. Citation analysis corroborates this pattern of late recognition: references to Simon (1955) rise gradually after publication, with citation peaks occurring only in the post-2000s period (see Figure 6).

However, Simon's influence remained limited in two critical respects. First, the slow diffusion of his framework meant that, by the time it achieved broad conceptual prominence, it faced direct competition from "new" behavioral economics, present in semantic clusters "Decision Theory" and "Decision Under Uncertainty", as well as in "Consumption, Savings & Intertemporal Choice" and "Game Theory". All of these four clusters expanded rapidly in the 1980s, with most of them individually surpassing cluster "Rationality & its Limits" in their share of sentences. Although these strands incorporated elements of bounded rationality, they did so primarily through the conceptual tools of "new" behavioral economics—ranging from prospect theory in the analysis of risk to experimental paradigms such as the ultimatum game in game theory (Guala 2008).

Second, Simon's conceptual influence, spearheaded by bounded rationality, never translated into stable research communities. While his framework shaped how many economists approached rationality, very few actually developed the research program he initiated. Bibliometrically, only small, unstable communities formed around Simon's work. In the 1960s and early 1970s, his ideas appeared in *Public Economic* clusters, operations research circles and behavioral theories of the firm (*cl_165*) and critique of the standard profit-maximizing such as socialist enterprise theory (*cl_123*). Simon was generally associated with other researchers critical of perfect rationality: Winter (1971), which applied satisficing to firm behavior; Cyert and George (1969) on behavioral theory of the firm; and Leibenstein (1966) on "X-efficiency," challenging economics' focus on allocative efficiency (*cl_179*). Together, Simon, Winter, Cyert, and Leibenstein formed a diverse organizational theory critique of how rationality, especially through profit maximization, is employed in economics. While this collective movement had some momentum, none of these research had an individual effect on economics structure (see also Sent 2004). After 1969-1976, these different research streams coalesced into small short-lived community (*cl_258*) before fragmenting. References to Simon and critiques of profit maximization remained scattered, moving through several small communities (e.g., *cl_793* on post-keynesian economics, *cl_755* on contract theory, *cl_773* and *cl_614* on institutionnalist approaches) without ever achieving the critical mass. While Simon's influence in economics was conceptually enduring, it remained marginal as a structuring force, unable to generate coherent, large and stable research communities.

"New" behavioral economics reception reveals a dramatic contrast. Unlike Simon's trajectory, the heuristics and biases research program of Kahneman, Tversky, and other "new" behavioral economists generated multiple stable clusters that formed well-identified and substantial research communities. A main "Behavioral Economics: Risk & Uncertainty" cluster emerged in 1981-1988, structured by early adopters of the new approach. The crystallization of this distinct cluster signaled the beginning of explosive growth and the spinning off of several autonomous communities that captured emerging specialized areas of behavioral economics: pro-social behavior (*Behavioral Economics: Social Preferences*) and behavioral game theory (*Game Theory: Behavioral*), intertemporal decision-making and risk (*Behavioral Economics: Risk & Uncertainty*), and behavioral finance (*Finance: Behavioral Finance*). In a lot of these cases, behavioral economics clusters come to outgrow or replace existing clusters (e.g., game theory, decision theory).

By the 2000s, behavioral economics had become the dominant venue for research on rationality. Three behavioral economics communities represented approximately 30% of the entire network with many additional communities structured by behavioral economics research without being explicitly characterized as such. The scale and speed of acceptance differed dramatically from Simon's experience. By 1980, Kahneman and Tversky (1979) had already surpassed Simon (1955) in annual citations (Figure 6). By 1985, prospect theory had already surpassed the lifetime citation peak that Simon's work would ever reach within economics. By the 2000s, most microeconomic publications addressing rationality were explicitly connected to the behavioral economics research program. While bounded and procedural rationality remained frequently invoked concepts, no subsequent research community clearly carried Simon's legacy as a stable bibliometric anchor.

The contrast is stark and raises fundamental questions about disciplinary reception. Our quantitative study yields two particularly surprising results regarding the contrasting influence of Kahneman and Simon. First, the temporal patterns of adoption differ radically. For the historian Heukelom (2014), a pivotal moment in the history of behavioral economics was the explicit creation of research programs within the Sloan and Russell Sage Foundations between 1984 and 1992, notably fostering the Kahneman–Thaler collaboration that decisively transformed the field's trajectory. Our analysis, however, suggests that a distinctive dynamic was already underway by the early 1980s in the reception of Kahneman and Tversky's work. Despite the fact that Kahneman and Tversky (1979) was the duo's only publication in an economics journal until 1985—well before the Sloan–Russell Sage programs formally institutionalized behavioral economics—prospect theory experienced rapid and sustained citation growth (Figure 6). This align with a general findings in scientometrics that citations in the first two years after publication explain more than half of the variation in cumulative citations (**sternHighRanked2014?**). While the institutionnal structures that came after played an important role in the subsequent reception of the research program as a whole, it is important to recognize how the reception of Kahneman and Tversky's differ greatly in the couple of years after publications. Moreover, as early as the 1978–1985 bibliometric network, we identify an autonomous cluster structured around their work (cl_391), indicating that the article not only attracted citations but quickly catalyzed new research strands, in a way Simon's work never did within economics (Heukelom 2012).

The second surprising finding concerns the relationship between "old" and "new" behavioral economics. Contrary to narratives of rupture or forgetting, citations to Simon (1955) have never been as high as during the rise of new behavioral economics—a paradoxical pattern given recurrent claims that the field has neglected its intellectual roots. Earl (2022) criticizes this apparent amnesia:

> Neither Kahneman nor Thaler have sought to promote earlier behavioral economics alongside more recent work. Instead, they give the impression that behavioral economics started around 1979–1980 with the publication of Kahneman and Tversky's (1979) article on prospect theory and that theory's use by Thaler (1980). All in all, this is a very curious state of affairs: a cynic might suggest that it looks rather as if the earlier work has been airbrushed from the history of economic thought by the strategic redefinition of what constitutes behavioral economics. A more charitable and reflexive view would see the situation as resulting from insufficient familiarity with the earlier literature (Earl 2022, 2)

The rising citations to earlier work by Simon and Allais demonstrate that, while "new" behavioral economists may not consistently acknowledge the contributions of "old" behavioral economics, the rapid expansion and growing scale of the field have nevertheless drawn unprecedented attention to these earlier works, far exceeding the recognition they received at the time of their original publication. This pattern admits at least three interpretations, each carrying distinct implications for how we understand the evolution of economic thought.

A favorable interpretation holds that "new" behavioral economics has genuinely revived previously abandoned research directions, moving toward a reconciliation with earlier strands—a trajectory explicitly advocated by Sent (2008) and more recently by Earl (2022). A more critical reading emphasizes intellectual appropriation, whereby classic references are selectively reframed to fit the new agenda, renewing interest but through a biased and presentist lens, as Mongin has argued in the case of Allais (Mongin 2019). A third interpretation suggests that rising citation counts reflect a growing backlash against the "new" behavioral program, as defenders of "old" behavioral economics increasingly invoke Simon and others to challenge the heuristics-and-biases framework.

Our findings tentatively support the second interpretation. The sparse and weakly structured citation patterns associated with "old" behavioral economics, combined with the fact that contemporary discussions of rationality are increasingly framed through the concepts and tools of "new" behavioral and experimental economics, point toward appropriation rather than genuine integration. Citations may acknowledge earlier work without engaging its distinctive methodological commitments or theoretical ambitions. Simon's influence thus persists primarily as a symbolic and framing reference, useful for situating debates and reconstructing intellectual lineages, while his substantive framework remains only superficially incorporated. Although a definitive assessment would require closer qualitative analysis of how these citations function in contemporary texts, our study provides a quantitative roadmap for future research on the complex and ambivalent relationship between "old" and "new" behavioral economics.

Conclusion

This article is first and foremost methodological: it aims to demonstrate, through a concrete application, the usefulness of quantitative methods for the history of economic thought. Nonetheless, it does so by breaking new ground in the history and philosophy of economics. First, the corpus we build—comprising nearly 290,000 full-text economics articles spanning more than a century—is, to our knowledge, the largest and most comprehensive database ever used to study economic contributions in English. Second, this article constitutes the first application of quantitative methods to the study of "semantic change" within the history of economics, an area that represents an important and growing strand of the literature on quantitative text analysis (Montanelli and Periti 2024).

Our approach allows us to identify, within this corpus, the sentences most closely associated with the words "rational" and "rationality" and, by extension, the articles most likely to engage with rationality in one way or another. By allowing the algorithms to identify dozens of semantic clusters and bibliometric communities within short temporal windows (eight years or a decade), and by then aggregating these groupings over time and characterizing them through multiple indicators, we can conduct analyses at different scales. This makes it possible both to *zoom out*, by examining the overall configuration of clusters across periods, and to *zoom in*, by focusing on subsets of clusters more directly concerned with rationality. As shown in the previous section, a panoramic view that considers all semantic clusters may initially include groups that appear only loosely connected to rationality—especially in the early decades. While examining these clusters can be informative for understanding the broader intellectual context, the analyst can also foreground more explicitly rationality-oriented clusters when addressing more targeted research questions. This flexibility is only possible because the initial corpus is sufficiently inclusive, reducing the risk of overlooking relevant developments. More generally, our approach allows the scale of analysis to be adjusted to the question at hand, rather than imposing it ex ante.

We want to conclude by drawing several general methodological and historiographic claims from the approaches developed in this article.

First, although quantitative methods in the history and philosophy of economics have been applied mainly to the postwar—and often post-1970s—period, our study shows that they can also be used effectively for earlier periods. Despite problems of text recognition in JSTOR articles prior to the 1930s, our methods nonetheless provide a fine-grained depiction of economic debates in the early twentieth century. In particular, we were able to recover most of the institutionalist contributions on rationality highlighted by Rutherford (2013) and Yonay (2001). At the same time, our results bring to light additional contributions not covered in these accounts, thereby opening avenues for further inquiry. Of course, some important works remain absent because they were published as books rather than journal articles. This limitation suggests that extending the corpus to include books would be a worthwhile direction for future research.

Second, a key property of groupings produced by unsupervised methods is that they bring together documents that share cognitive content without presupposing agreement. The texts clustered together address the same objects or problems, but they may do so in divergent, and sometimes opposing, ways. As a result, our methods make it possible to identify sites of resistance to dominant models or ideas. What emerges most clearly are the objects that attract sustained attention; within those sites, one can then examine who endorses, revises, or contests particular ways of treating them. This feature opens the way to a more "symmetrical" historical analysis (Bloor 1976), one that gives analytical weight not only to intellectual "winners" but also to those whose positions did not ultimately prevail.

Finally, these methods are also "discovery tools." While they can be used to confirm or challenge existing claims in the history and philosophy of economics, they also help identify patterns that have so far been absent—or only marginally present—in the historical literature. We have briefly pointed to some of these patterns above. By making available an interactive application to explore our results, we hope to encourage readers to pursue their own paths of discovery within the corpus.

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Table 1.: Sentences from the corpus closest to the representative vector in 1910, 1950, and 2000.

Sentence	Cosine similarity
1910	
From another side, the concepts supply the basis for rationality.	0.690
The recital is important only as tending to show that a theory of valuation which places the emphasis upon rationalistic appraisal overlooks the most important features of the process which it seeks to explain.	0.686
Complete reliance may be placed upon the rationality of both the pecuniary and the hedonic subject.	0.678
It is exceedingly doubtful whether such hyper-rational view of human nature could have gained wide credence among men not themselves disciplined in the use of pecuniary concepts.	0.670
More fundamental is the great problem of accounting for economic rationality itself.	0.656
1950	
For rationality of choice is nothing less than the choice with complete awareness of the alternative rejected.31 Rationality is connected in economic thought with maximization.	0.777
One must be sensible about rationality.	0.750
We have to assume that man is rational and that his rationalism leads him to act as industrial civilization requires him to act if we want to be able to live under the present economic system.	0.728
For Lionel Robbins the postulation of intrinsic rationality as the norm which relates means to ends has a firm empirical basis.	0.726
Rational behavior means conduct motivated by the goal of maximization of money income, wealth and economic advantages.	0.718
2000	
Our aim was to provide on overall picture how decisions of boundedly rational decision makers emerge.	0.808
An important approach to discussing the notion of rationality may be found in .	0.803
Lack of agreement on the definition and operational import of bounded rationality is a major obstacle (Ariel Rubinstein 1998; David Kreps 1999).	0.803
For further discussion of the informal rationality constraint's connection to decision-theoretic conceptions of rationality, see Carroll (1999).	0.803
Much economic research is aimed at the conditions under which certain types of rationality are dominating, and what types of measures are feasible to affect this domination.	0.802

Figures

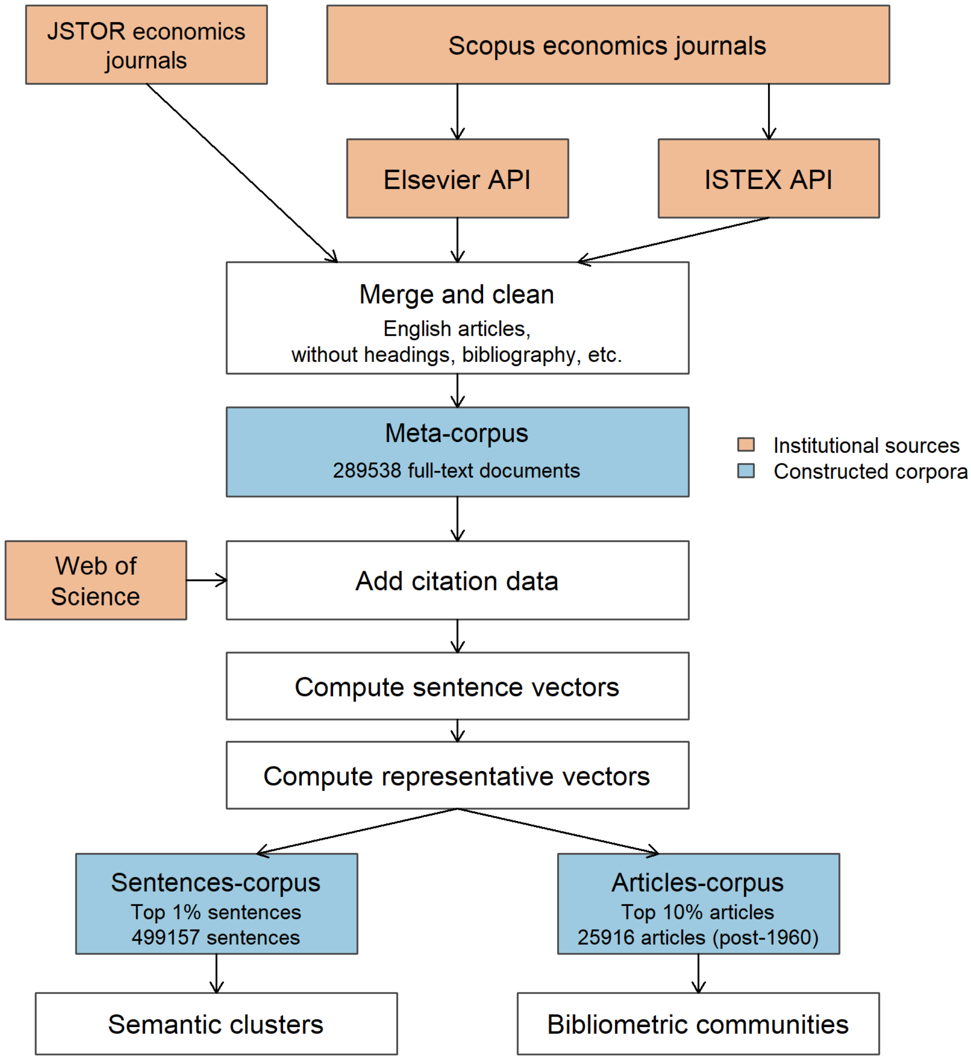


Figure 1.: General workflow of the methods used in the paper. Orange boxes represent data sources. Blue boxes represent used corpora. White boxes are transformation steps. Arrows indicate the flow of data and transformations. Table 1 and Table 2 give examples of sentences and documents in the sentences and documents corpora.

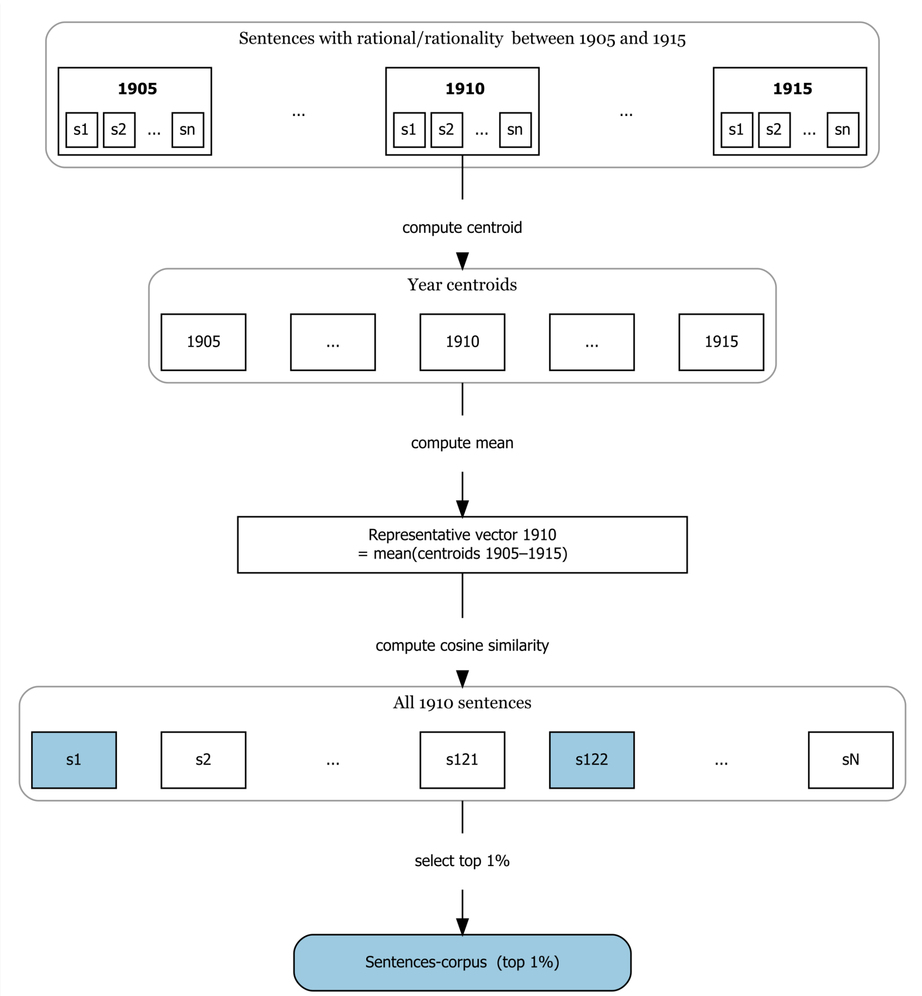
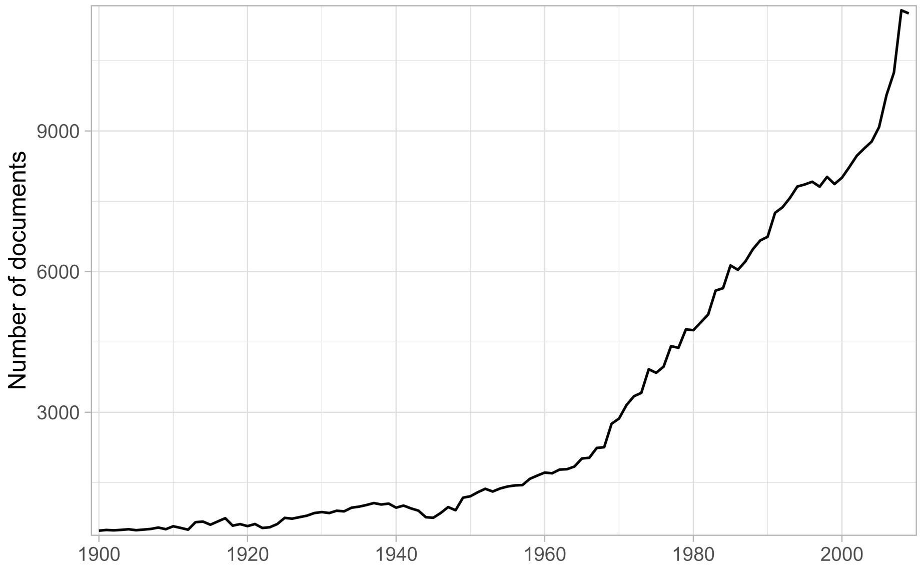
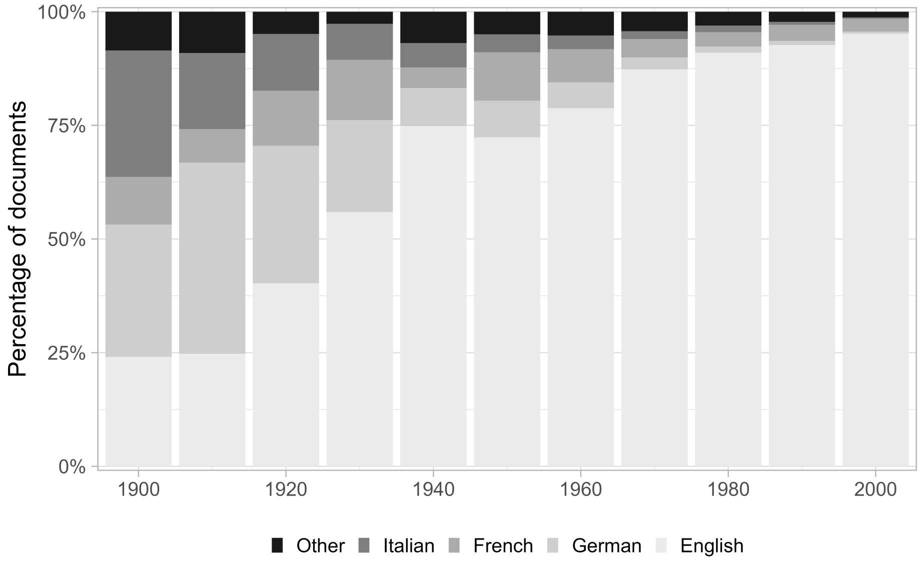


Figure 2.: Zoom on the construction of the representative vector of rationality and how it is used to build a corpus (here the sentences-corpus). White boxes are transformation steps. Arrows indicate the flow of data and transformations. Table 1 and Table 2 give examples of sentences and documents in the sentences and documents corpora.



(a) Documents



(b) Languages

Figure 3.: Distribution of documents and languages in the meta-corpus over time (1900-2009).

Table 2.: Documents from the corpus closest to the representative vector in 1910, 1950, and 2000.

Title	Authors	Journal	Cosine similarity
1910			
The Rationality of Economic Activity	Wesley C. Mitchell	Journal of Political Economy	0.862
Social Productivity Versus Private Acquisition	H. J. Davenport	The Quarterly Journal of Economics	0.824
The Futility of Marginal Utility	E. H. Downey	Journal of Political Economy	0.786
Observation in Economics. Annual Address of the President	Davis R. Dewey	American Economic Association Quarterly	0.778
The Economics of Henry George's "Progress and Poverty"	Edgar H. Johnson	Journal of Political Economy	0.765
1950			
The Economic Way of Thinking	Howard S. Ellis	The American Economic Review	0.843
What Kind of Psychology Does Economics Need?	C. Reinold Noyes	The Canadian Journal of Economics and Political Science / Revue canadienne d'Economique et de Science politique	0.815
Government Economic Policy: Scope and Principles	W. A. Mackintosh	The Canadian Journal of Economics and Political Science / Revue canadienne d'Economique et de Science politique	0.810
Psychological Assumptions of Economic Theory	Albert Lauterbach	The American Journal of Economics and Sociology	0.810
Economics and Value Judgments	Paul Streeten	The Quarterly Journal of Economics	0.809
2000			
Rational Irrationality: A Framework for the Neoclassical-Behavioral Debate	Bryan Caplan	Eastern Economic Journal	0.867
Darwinian selection does not eliminate irrational traders	Biais B.	European Economic Review	0.826
Rationality, Imagination and Intelligence: Some Boundaries in Human Decision-making	Augier M.	Industrial and Corporate Change	0.816
Costly and Bounded Rationality in Individual and Team Decision-making	Radner R.	Industrial and Corporate Change	0.810
Alternative models of individual behaviour and implications for environmental policy	Van Den Bergh J.C.J.M.	Ecological Economics	0.797

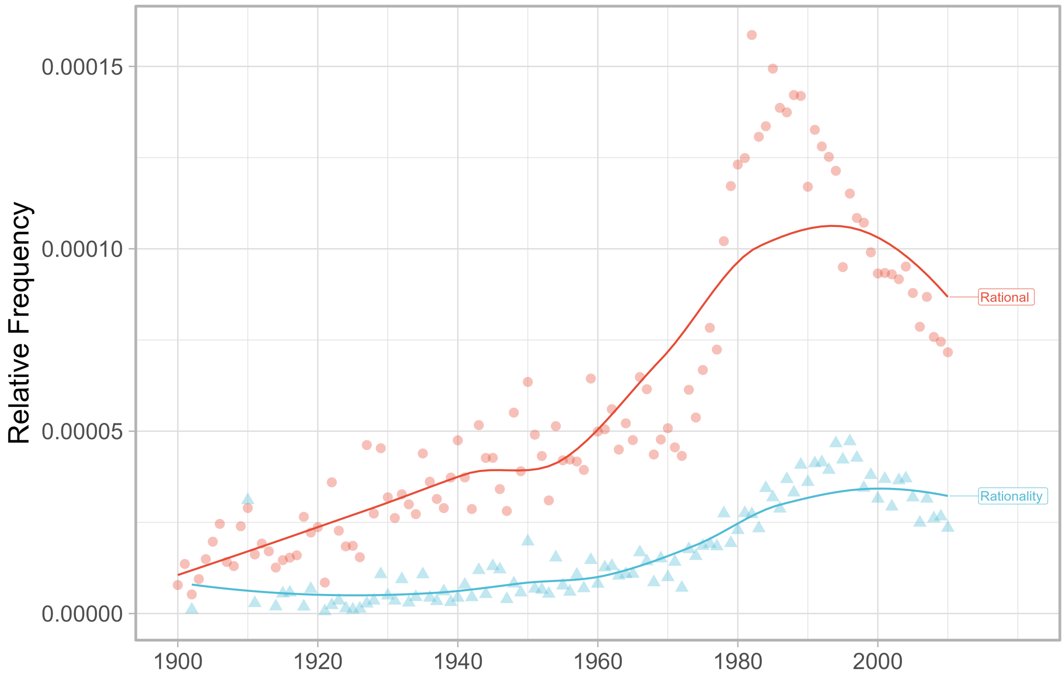


Figure 4.: Relative frequency of terms "rational" and "rationality" in the jstor database (1900-2009). Points are the empirical data. Lines are the loess smoothed trends.

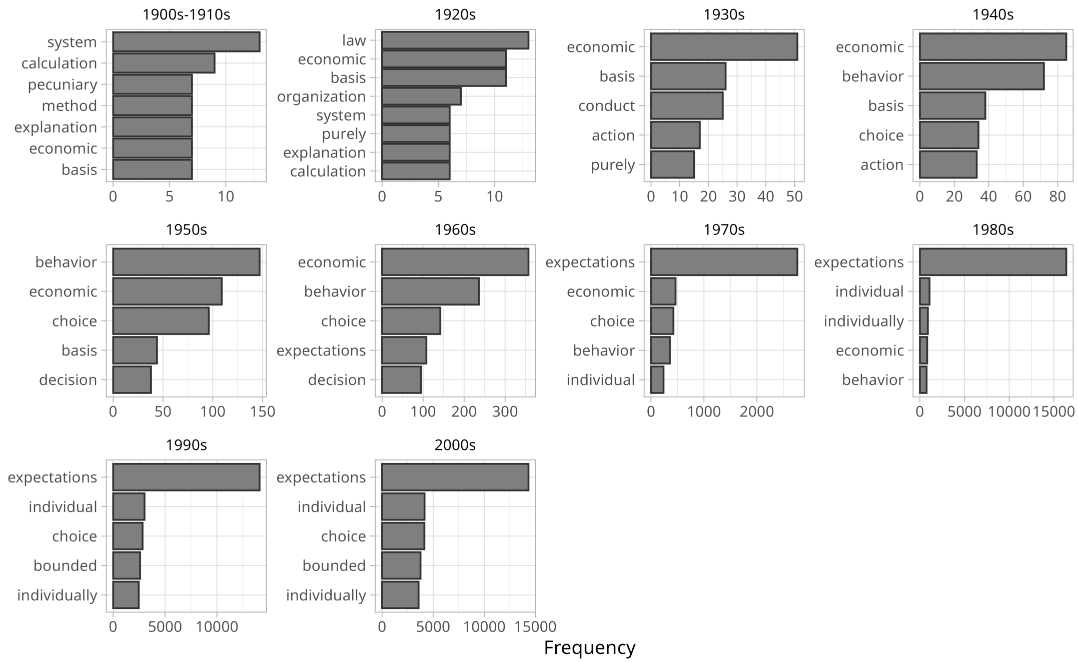


Figure 5.: Five most frequent neighbor terms of "rational" and "rationality" by decade. We search and select $n-1$ and $n+1$ neighbors of "rational" and "rationality" in each sentence. We remove stop words and compute the most frequent neighbors by decade. We merge 1900 and 1910 decades because of the low number of sentences containing "rational" or "rationality" before 1910.

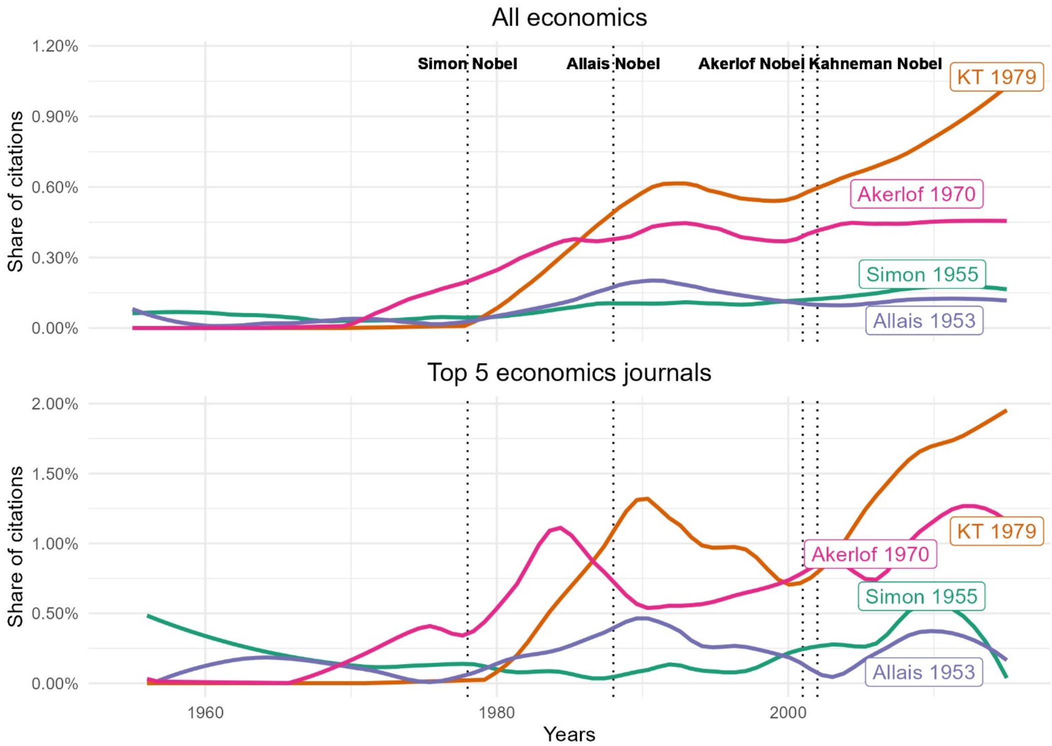


Figure 6.: Relative citation of a selection of seminal papers on behavioral economics.

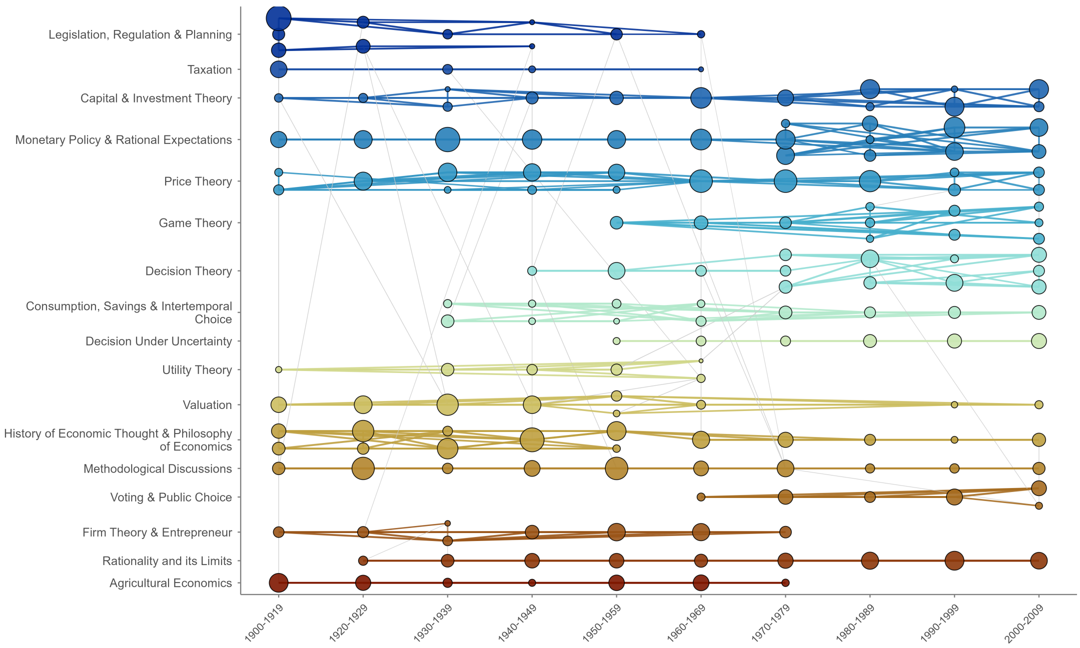


Figure 7.: Each node is a group of sentences discussing rationality in a similar way, identified within a given decade. Node size reflects how many sentences belong to that group. Colors indicate broader thematic clusters that persist across decades — nodes sharing the same color address rationality in comparable ways across time. See the appendix for the full methodology. In this visualisation, we group clusters by decade along the x-axis (1900–1919, 1920–1929, ..., 2000–2009) and group thematically similar clusters along the y-axis, making it easier to follow how topics evolve over time. We strongly advise readers to explore all time windows interactively via the companion application.

Figure 8.: Each node is an article from the 1994–2001 time window. Articles are connected when they cite similar references—a proxy for shared intellectual background. Colors indicate communities of articles that recurrently cluster together across consecutive 8-year time windows. Uncolored nodes belong to communities that are too small or too short-lived to be considered substantive. The spatial layout positions articles with similar citation profiles in close proximity. See the appendix for the full methodology. We strongly advise readers to explore all time windows interactively via the companion application.

Appendix

Figure 1 summarizes the general workflow of our method. The goal of this appendix is to provide the general rationale and workflow of our methods. Since fulltexts are not publicly available, we cannot share a full replication package and, in particular, the data and the code to build the meta-corpus are not available. The rest of the code — building the sentences and documents corpora, building the representative vector, clustering methods, bibliometric methods — is available in the GitHub repository:
https://github.com/agoutsmedt/ejhet_quanti_method. We refer to the companion application for the detailed results of the clustering analyses. A public access is provide at the following link: https://019adac8-81d4-aa0e-808c-08861c261fd2.share.connect.posit.cloud/.

Building the meta-corpus

The meta-corpus is built from three main sources: JSTOR, Elsevier and ISTEX. The main source is JSTOR, which provides the largest coverage of historical economics journals. However, some important economics journals published by Elsevier are missing from JSTOR. We therefore complement JSTOR with documents from SCOPUS journals using two supplementary sources: Elsevier and ISTEX. We used Elsevier API to access Elsevier's economics journals full-texts after 1997. We used ISTEX because Elsevier documents were not available via the Elsevier API before 1997.

To identify economics documents, we use a journal-based approach. We build a list of economics journals and then retrieve all documents published by these journals in our sources. To build this list, we first retrieve all journals classified as economics in our sources and in the Econlit database—a bibliographic database of economics literature maintained by the American Economic Association. We review them manually to ensure they are (i) economic oriented journals and (ii) peer-reviewed journals. We also filtered out very recent journals (less than 20 years old). The rationale was to limit the short-time effect of small journals and focus on journals that have a long-term influence on the field. We end up with 316 unique journals. The list of journals is available in the supplementary material.

We then queried each source to retrieve all available documents published by these journals between 1900 and 2009. We keep only documents classified as research articles (i.e., we remove editorials, book reviews, etc.) and written in English. This results in a total of 289,538 documents that represent our curated data. Table 3 summarizes the sources of our meta-corpus. Each source provides documents at different levels of granularity: Elsevier provides paragraph-level data, JSTOR page-level data, while ISTEX provides no such segmentation. We thus paste together the full-texts from each source at the document level to harmonize the unit of analysis.

The full-texts from JSTOR and ISTEX are obtained through OCR, which introduces noise in the text. In contrast, Elsevier provides structured full-texts with minimal noise. We apply basic cleaning steps to the OCR-based full-texts based on RegEx patterns to remove paratextual material (e.g., affiliations, acknowledgements, headers, references) that introduce important noise. However, cleaning full-texts at scale is computationally intensive and not easily scalable and some noise is likely to remain in the data.

To perform our bibliometric analysis, we enrich our fulltexts with citations from Web of Science (WOS). We match our fulltexts database and WOS data based on their metadata (e.g., journal, year, volume, first page). We use metadata-based matching rather than DOIs because many documents, especially older ones, do not have DOIs. Because metadata might slightly differ across our sources, we also applied combinations of metadata rather than exact matches on all metadata fields. Our algorithm also includes fuzzy matching to account for OCR errors in titles and journals. We achieve an overall match rate of 70.1% across the full database. Column "WOS matched (%)" in Table 3 indicates the proportion of documents successfully matched to WOS records for each source and each decade. Table 4 details the WOS match rate by decade. Coverage is near zero before 1940, as WOS records are sparse for this period. From the 1950s onwards, match rates stabilise between 60% and 68%.

Building the sentences and documents corpora

Cleaning sentences before computation

In order to compute sentence embeddings, we first need to split the full-texts into sentences. We use `sent_tokenize` from the NLTK Python library. While there are more advanced models for sentence segmentation (e.g., `spacy`), they are computationally intensive and not easily scalable to a corpus of millions of sentences.

Sentence segmentation may amplify noise in OCR-based texts from JSTOR and ISTEX, which often contain punctuation and capitalization errors. Given the size of the corpus, errors from the OCR or sentence segmentation cannot be fully removed at scale. We only remove sentences for which the segmentation is almost certainly incorrect and provide no information, i.e., sentences that contain less than 20 characters, less than three words, or contain more than 50% digits. Remaining errors are expected to be averaged out by the clustering analysis.

A more serious issue is the presence of paratextual sentences (e.g., affiliations, acknowledgements, headers, references) in JSTOR and ISTEX OCR full-texts. These sentences introduce systematic noise that can affect clustering analysis. For instance, because references often contain similar wording (e.g., "Journal of…", "Vol.", "pp.", etc.), they tend to cluster together in the embedding space. We use this property at our advantage to remove such noise. Because paratextual sentences tend to be semantically similar, they form dense regions in the embedding space, which we exploit to identify and remove them. We first define four categories of noisy sentences: affiliations, acknowledgements, headers, references. For each category, we use regular expressions to identify a set of sentences that likely belong to this category. We compute (i) the centroid vector of these noisy sentences and (ii) the cosine similarity between those centroids and each sentence embeddings from our meta-corpus. We remove sentences from our corpus that are above a certain similarity threshold with any of the four noisy centroids. We determine the threshold using Youden's J statistic and by manually inspect sentences around the threshold. We exclude sentences flagged as noise from all subsequent analyses.

Compute sentence embeddings

We compute sentence embeddings with Sentence-BERT, using the pretrained model `sentence-transformers/all-mpnet-base-v2`. This model is widely used in natural language processing applications and has been shown to perform well for semantic similarity tasks. We use the `sentence-transformers` Python library to compute the embeddings. This model yields a single 768-dimensional vector per sentence. Let's denote the set of sentences in our meta-corpus as $\mathcal{S}$. For each sentence $s \in \mathcal{S}$, we compute its embedding $v_s \in \mathbb{R}^{768}$ using the Sentence-BERT model.

Compute the representative vectors

The computation of the representative vectors is summarized in Figure 2. Starting from the embedded sentences of the meta-corpus, we first keep the sentences whose text matches "rational" or "rationality".[37] Then, for each year, we compute the centroid of these

[37] We filter the text using the RegEx: `\brational(?:ity)?\b`.

embeddings to obtain a year-specific embedding. We then compute an unweighted centered moving average over an 11-year window (year $t \pm 5$) to produce the final representative vector series used in subsequent similarity calculations. We rely on an unweighted average to give equal importance to all years rather than giving more to years with more sentences.

Let $\mathcal{S}_t^R$ denote the set of sentences in year t that contain "rational" or "rationality". We compute the centroid $\bar{v}_t^R$ of the embeddings of these sentences as follows:

$$\bar{v}_t^R = \frac{1}{|\mathcal{S}_t^R|} \sum_{s \in \mathcal{S}_t^R} v_s^R$$

We define the representative vector for year t, denoted as RV_t, as the unweighted average of the centroids of the 11-year window centered on t:

$$RV_t = \frac{1}{11} \sum_{i=-5}^{5} \bar{v}_{t+i}^R$$

Building the sentences-corpus and the articles-corpus

To build the sentences-corpus, we select sentences that are semantically close to the representative vector of their year, in terms of cosine similarity. For each year t, we thus compute the cosine similarity between the vector of sentences in $\mathcal{S}_t$ and the representative vector RV_t. For each year t, we then retain the top 1% most similar sentences. We thus obtain, the sentences-corpus, a set of sentence $\mathcal{S}_t^{RV}$ for each year t that are the most semantically similar to the representative vector of their year. *Importantly, $\mathcal{S}_t^{RV}$ and $\mathcal{S}_t^R$ are distinct sets.* The sentences in $\mathcal{S}_t^{RV}$ are likely to contain "rational" or "rationality" but this is not a requirement.

Similarly, to build the articles-corpus, we select documents that are semantically close to the representative vector of their year, in terms of cosine similarity. Because v_s is a sentence-level embedding, we first compute a document-level embedding by averaging the embeddings of all sentences in the document. Let's denote $\mathcal{D}$ as the set of documents d in our meta-corpus, and $\mathcal{S}_d$ the set of sentences in document d. We compute the embedding of each document d as follows:

$$v_d = \frac{1}{|\mathcal{S}_d|} \sum_{s \in \mathcal{S}_d} v_s$$

This results in v_d, a single 768-dimensional vector for each document. This vector represents the overall semantic content of the document. We then compute the cosine similarity between every v_d document in year t and the representative vector RV_t. For each year, we retain the top 10% most similar documents. This results is the articles-corpus, a set of documents $\mathcal{D}_t^{RV}$ for each year t that are the most semantically similar to the representative vector of their year.

The 1% and 10% threshold are of course debatable. It balances competing objectives. On the one hand, we want to capture a sufficient number of sentences and documents to represent the diversity of ways rationality is discussed in each year. On the other hand, we want to focus on sentences and documents that are strongly related to rationality and we want to limit the computational cost of subsequent analyses. There is clear room for improvement to choose less arbitrary thresholds.[38]

Overlap between sentences and documents corpora

Table 5 reports the size of $\mathcal{S}_t^{RV}$, the sentence-corpus, and D_t^{RV} the articles-corpus, as well as the extent of their overlap. By overlap, we mean the number of articles that are in both corpora. An article is in the sentences-corpus if at least one of its sentences is in $\mathcal{S}_t^{RV}$ and an article is in the articles-corpus if it is in D_t^{RV}. The sentences-corpus and the articles-corpus share 24,356 articles. Almost all documents in the articles-corpus are also in the sentences-corpus (i.e., 94%) but documents with sentences in the sentences-corpus are much less likely to be in the articles-corpus (i.e., 22.9%). This is expected given the different selection criteria: the sentences-corpus are drawn from any documents, not only the 10% most similar to the representative vector. However, despite representing only 22.9% of the articles in the sentences-corpus, these shared articles account for 51.7% of its sentences. In other words, sentences from articles in the article-corpus are over-represented in the sentences-corpus.

Semantic clusters

This section details the text clustering methods used on the sentences-corpus, $\mathcal{S}_t^{RV}$. Figure 9 summarizes the steps schematically. We describe each step in detail. Regarding the choice of clustering methods, we followed the workflow of Alammar and Grootendorst (2024) and used UMAP and HDBSCAN. However, we adopt a dynamic approach: we apply HDBSCAN for each decade and then we link clusters across decades to recover intertemporal semantic clusters. This is a common approach in the literature on semantic shift since most corpora are growing over time (Montanelli and Periti 2024). Applying a clustering algorithm on the whole corpus would lead to clusters that are mostly driven by recent years.

UMAP and HDBSCAN

UMAP is a dimensionality reduction technique that project the high-dimensional sentence embeddings into a lower-dimensional space while preserving their information. Intuitively, the rationale for using UMAP is that high-dimensional spaces are often sparse, making clustering algorithms like HDBSCAN less effective. Dimensionality reduction techniques like UMAP are standard practice to improve the performance of clustering algorithms (e.g., Benz et al. 2025). We first project the 768-dimensional sentence embeddings to a 100-dimensional UMAP space (metric = cosine, n_neighbors = 15, min_dist = 0.0). From each $v_s \in \mathbb{R}^{768}$ and $s \in \mathcal{S}_t^{RV}$, we obtain $u_s \in \mathbb{R}^{100}$. To fit UMAP, we use a sampling strategy that keeps all pre-1980 sentences and caps the post-1980 years at 30,000 sentences each. The rationale is to avoid over-representing recent years in the UMAP fit.

We then apply HDBSCAN to identify clusters in the UMAP space for each decade.[39] For each window, we run HDBSCAN on the UMAP coordinates. HDBSCAN's hyperparameters, in particular `min_cluster_size` and `min_samples`, influence the granularity of the clusters. We set `min_cluster_size = max(20, 1% of total sentences in the window)` and `min_samples = 1`. These parameters were selected after manual inspection of cluster interpretability. Figure 10 shows the distribution of noisy vs clustered sentences and the number of clusters identified per time window by HDBSCAN.

[38]For instance, defining a null model to determine a similarity threshold above which sentences or documents are significantly more similar to the representative vector than expected by chance.

[39]We define a first time windows of two decades (1900–1919)—because of the low number of sentences before 1920—followed by standard decades until 2009.

Merging clusters over time

We choose a network-based approach to link clusters across decades. This approach handles many-to-many relationships between decade-level clusters— a semantic cluster may gradually merge with or split into multiple clusters over time. It is also consistent with the network-based approach applied to the articles-corpus.

We represent each cluster identified by HDBSCAN as a node in a network and edges are the cosine similarity between all cluster centroids. This gives us a weighted network where nodes are decade specific clusters and edges are cosine similarities between their centroids. We keep only the most significant links using a backbone extraction (Domagalski, Neal, and Sagan 2021), using the LANS (Local Adaptive Network Sparsification, `alpha = 0.02`). LANS retains edges that represent statistically significant similarities relative to a null model in which the weights are randomly distributed among the node's edges. We apply the Leiden community detection (resolution = 2) on the backbone network. Intuitively, clusters that are strongly connected in the backbone network are grouped into intertemporal meta-clusters. Figure 7 is an example of the resulting network.

Bibliometric data and methods

This section details the bibliographic coupling and community detection workflow. Figure 11 summarizes the steps schematically. We describe each step in detail.

Dynamic coupling networks

Networks were constructed using bibliographic coupling, where two publications are linked if they cite common references. This approach identifies articles that draw on similar intellectual foundations, revealing thematic communities within the literature. Raw common references counting can be misleading, as articles that cite many references overall are more likely to share citations by chance and references that are highly cited are also overall more likely to link to articles by chance. To weight shared references between two articles, we use the coupling strength method from (Shen et al. 2019). Coupling strength weighs both the size of the reference list of articles and the number of overall citations of each reference to overweight significant co-occurrences of references.[40]

To study the evolution of networks over time, we employed a dynamic windowing approach:

- Time window: 8 years
- Overlapping windows: windows overlap to allow smooth temporal transitions (1980-1987, 1981-1988, etc.)
- Edge threshold: articles are only connected if they have at least 2 shared references

For each time window, we removed isolated articles (articles with no connections) and small components (connected groups of articles disconnected from the main network) ensuring that the analysis focused on the coherent and relevant literature.

Clustering algorithm, spatilization, and intertemporal analysis

Communities within each network were identified using the Leiden algorithm with modularity optimization (Traag, Waltman, and Eck 2019). This algorithm identify groups of articles that are have strong connections to each others, but few connections to the rest of the networks (i.e., relevant sub-groups of articles across the main network).

[40]Statistical determination of edges are best options but are computionally unfeasable given the size of our networks

To identify the most relevant clusters across multiple networks, we made to an intertemporal analysis of our cluster. To trace the evolution of research communities over time, we linked clusters across consecutive time windows. Communities were considered continuous if they shared at least 50.1% of their members across two time windows. For example, if a cluster in the 1990-1997 network is made of similar articles as another cluster in the 1991-1998 network, the cluster are identified as the same one. This approach allows us to follow how research communities emerge, grow, split, merge, or decline over the 50-year period.

For visualization and interpretation, see the application.[41] We focused on substantive communities using two criteria:

- Temporal persistence: Communities must appear in at least 2 time windows
- Size significance: Communities must represent at least 5% of the network in their peak window

Smaller or more transient clusters are uncolored in visualizations to emphasize the major research streams.

To visualize individual networks, articles (nodes) are color-coded by intertemporal clusters and spatiliazed in a two-dimensional space using the Force Atlas 2 algorithm (Jacomy et al. 2014). This layout enhances readability by positioning similar articles in close proximity. However, these visualizations are intended as heuristic aids rather than primary analytical tools. The study's findings are grounded in quantitative network metrics and detailed tables rather than visual analysis.

Characterizing Research Communities

As a way to have a first exploratory labels for clusters, names were generated using a local open large language model (Gemma 3 27B) through the Ollama framework. For each community, we provided the model with:

- The 20 most frequently cited references (with titles when available)
- The 20 most productive authors
- The 20 highest TF-IDF terms

The model was prompted to generate concise labels capturing the thematic focus of each cluster. Note that these labels are not definitive but serve as starting points for interpretation.

[41] https://019adac8-81d4-aa0e-808c-08861c261fd2.share.connect.posit.cloud/

Table 3.: Sources of the meta-corpus (1900–2009).

Source	Documents	Journals	Unit of text	Format	WOS matched (%)
Elsevier	63111	122	paragraph	XML	78.1
ISTEX	60242	120	full text	raw OCR	69.6
JSTOR	166185	173	page	raw OCR	67.2

Table 4.: WOS match rate by decade in the meta-corpus (1940–2009).

Decade	Articles	WOS matched	Match rate
1940s	6,927	2,319	33.5%
1950s	10,193	6,946	68.1%
1960s	15,833	10,564	66.7%
1970s	33,162	23,377	70.5%
1980s	52,290	38,885	74.4%
1990s	70,641	51,564	73%
2000s	89,663	69,213	77.2%

Appendix Tables and Figures

Table 5.: Overlap between the sentences-corpus and the articles-corpus.

Corpus	Articles	Sentences	Shared articles
Sentences-corpus (top 1%)	106,376	499,157	24,356 (22.9%)
Articles-corpus (top 10%)	25,916	—	24,356 (94%)
Intersection	24,356	258,242	—

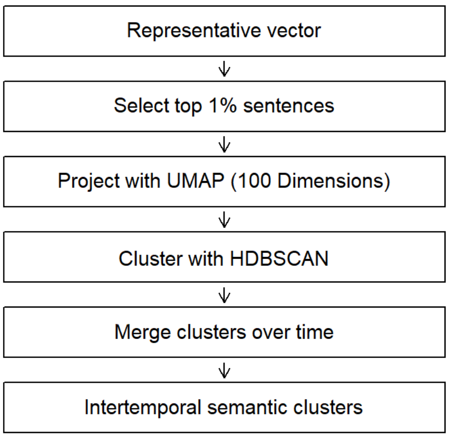
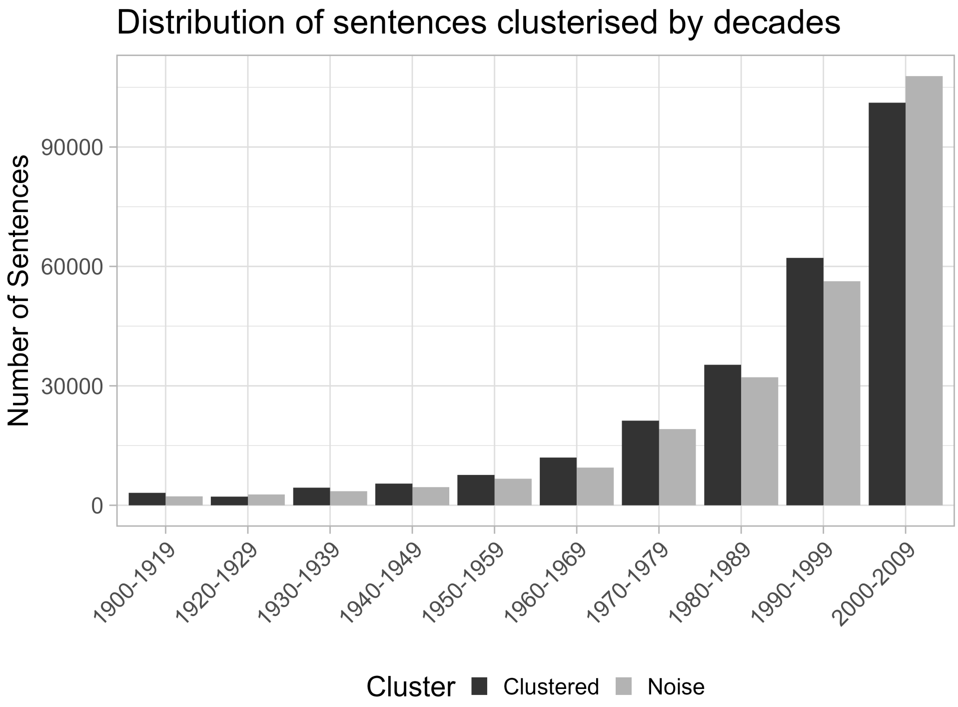
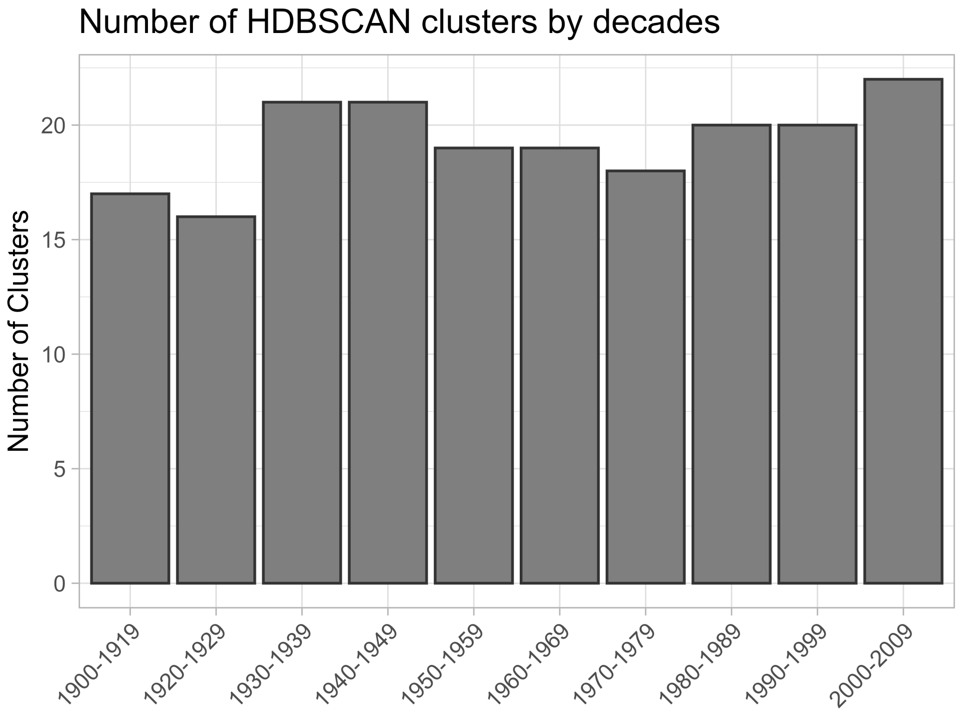


Figure 9.: Text clustering workflow.



(a) Noisy and clustered sentences



(b) Number of clusters by period

Figure 10.: Distribution of noisy vs clustered sentences and number of clusters per time window.

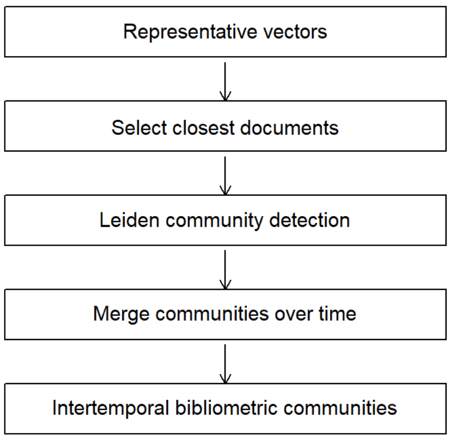


Figure 11.: Bibliographic coupling and community detection workflow.